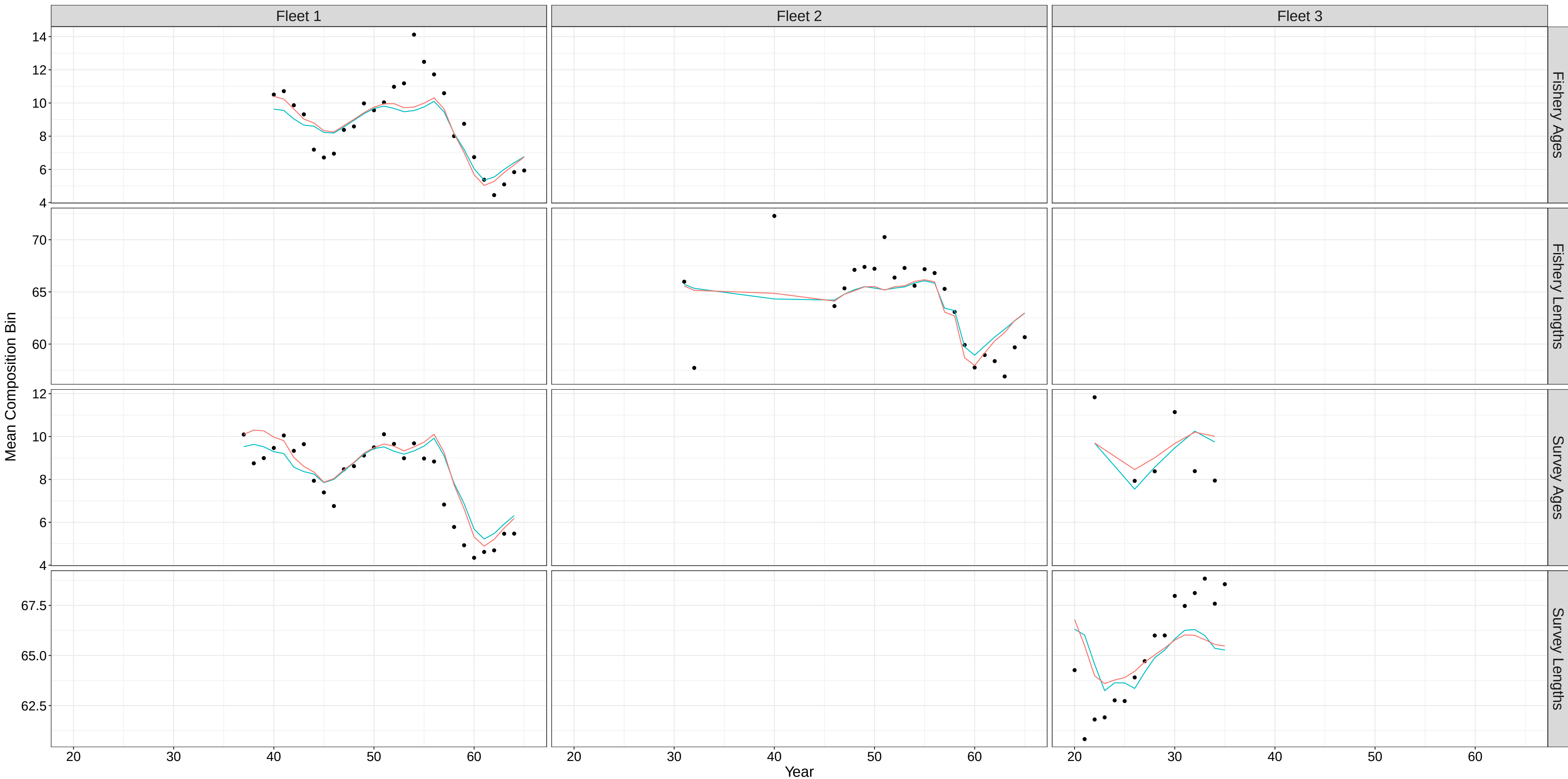


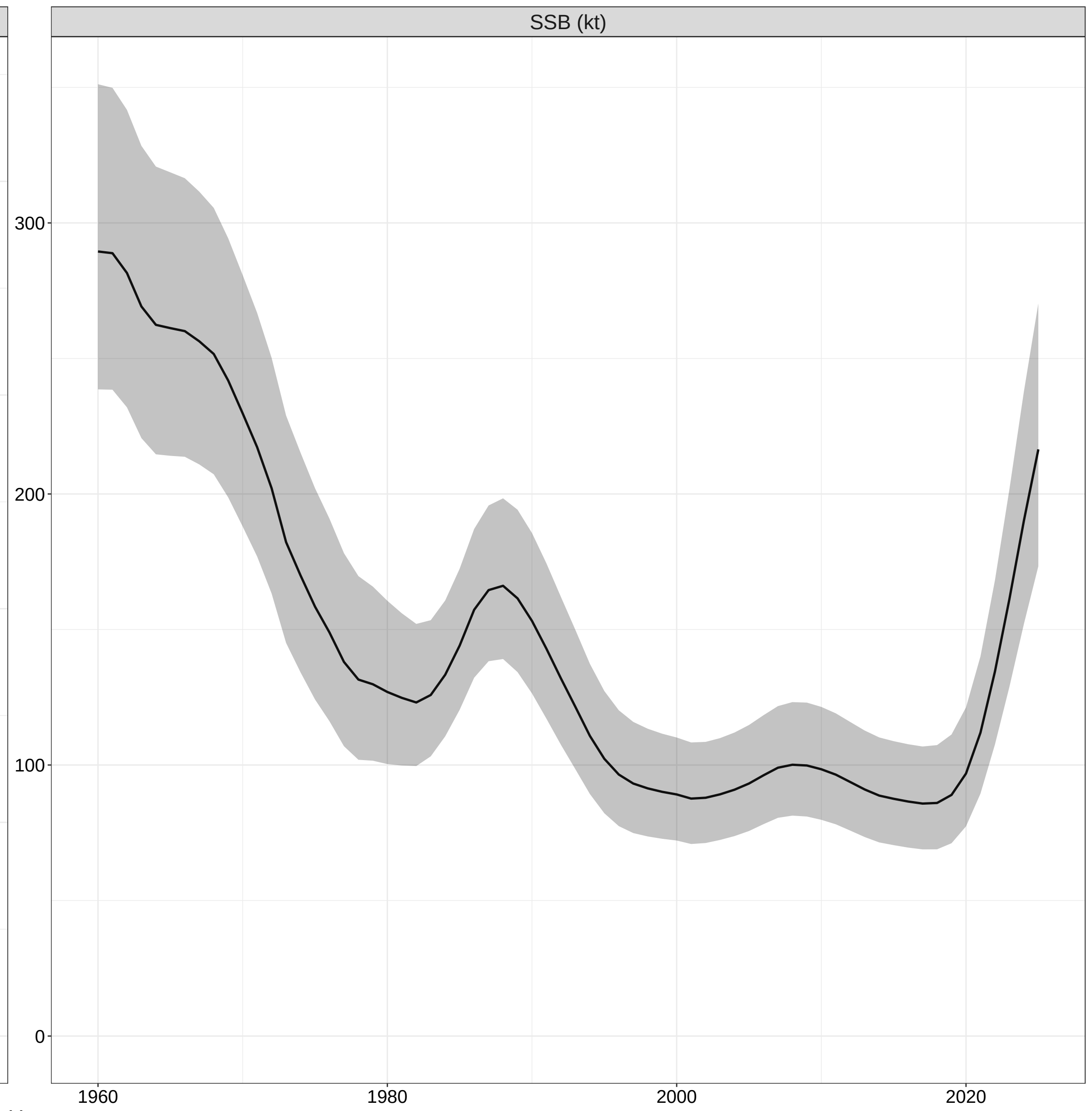
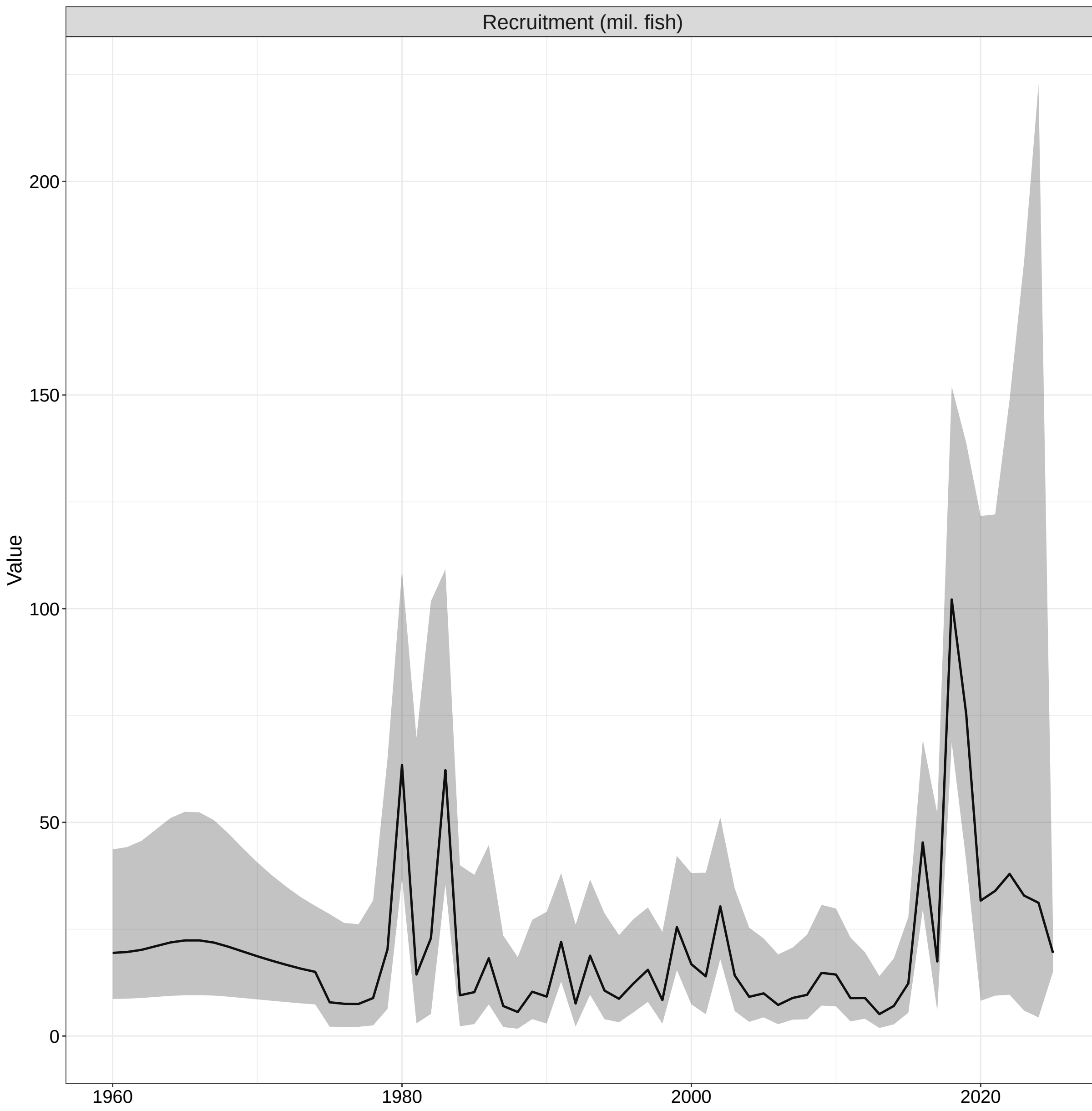
Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	0.07	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.12	145.15	17.69
LL Survey Ages	NA	NA	0.28	171.43	47.31
JPN LL Survey Ages	NA	NA	0.17	83.33	14.45
Trawl Fish Lengths	NA	NA	0.25	45.39	11.51
JPN LL Survey Lengths	NA	NA	0.34	54.62	18.84

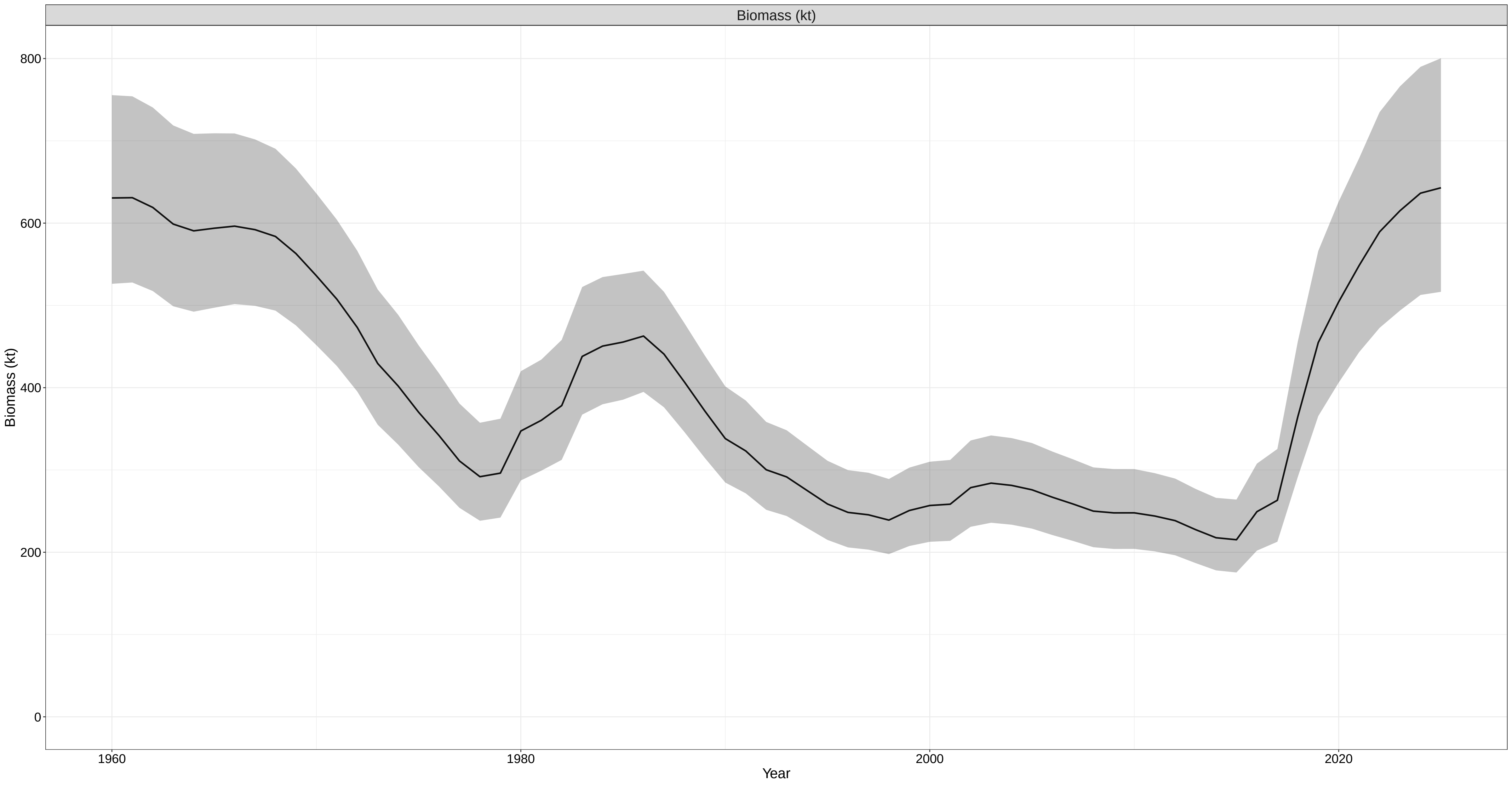
G3;Successfully passed post-optim-sanity checks

colour — First Iter — Last Iter



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	216.45211	369.66892	0.0464291278338114	40.85173	139.01719	0.07903	1.55702	0.58553	0.58749

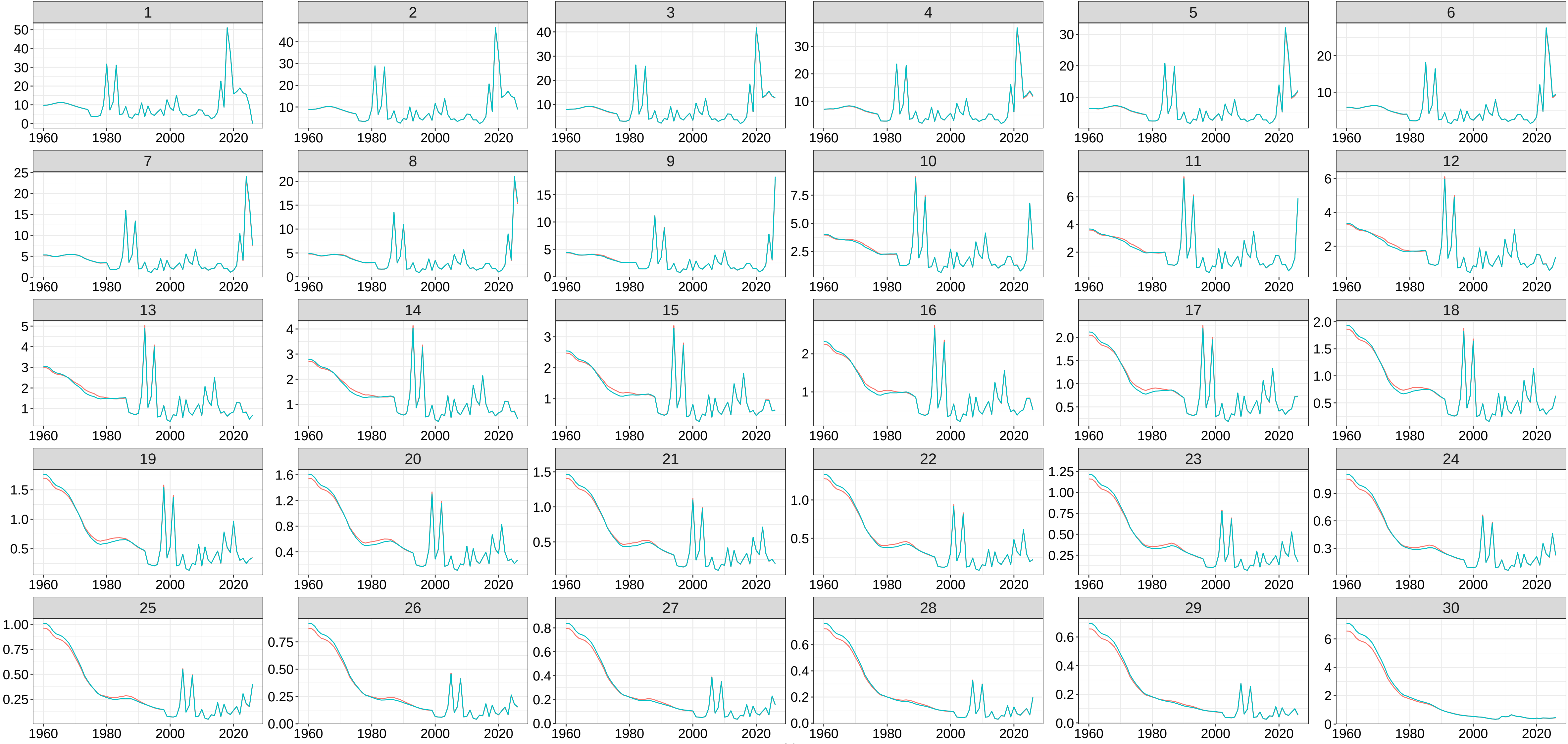


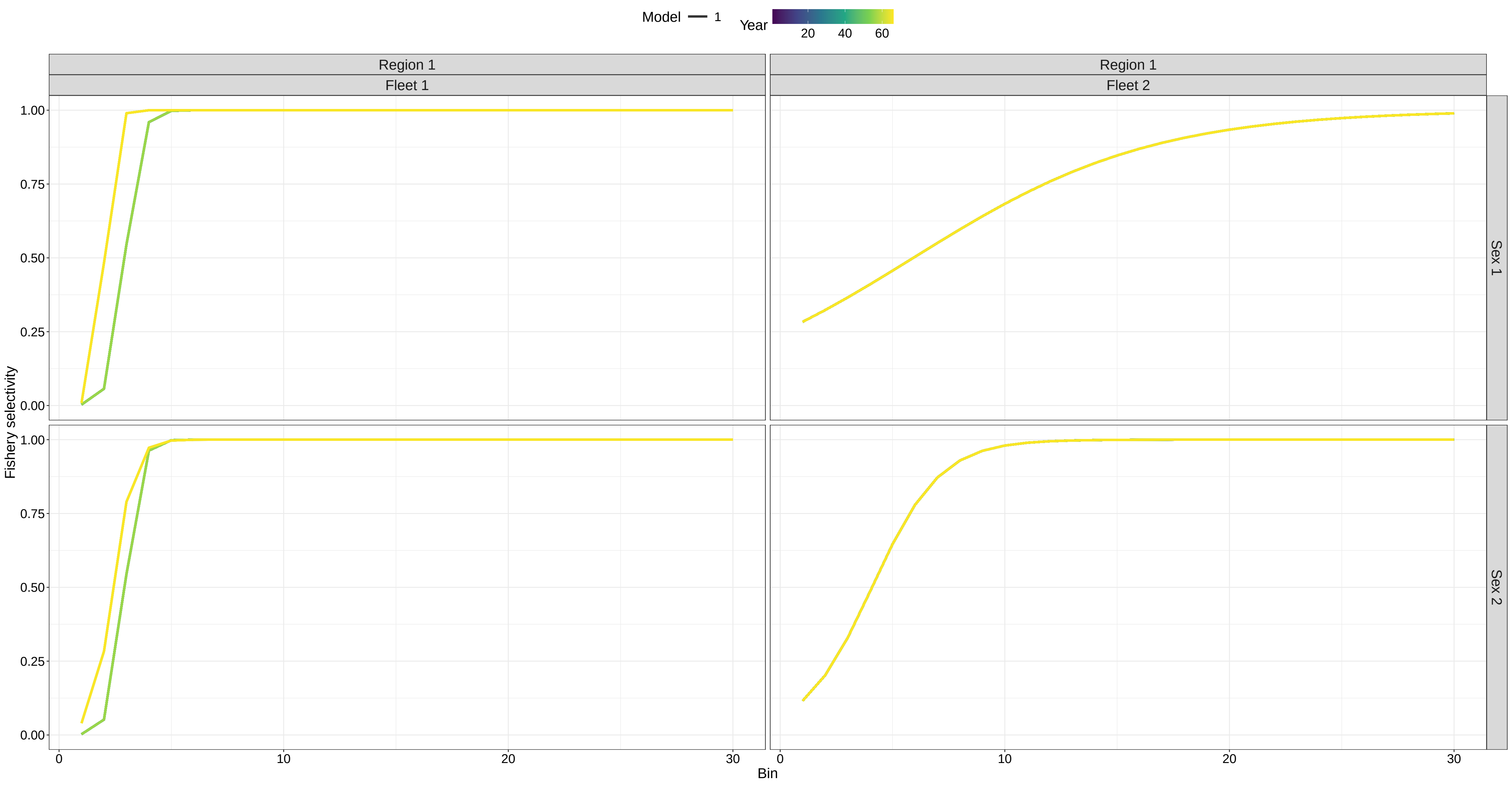


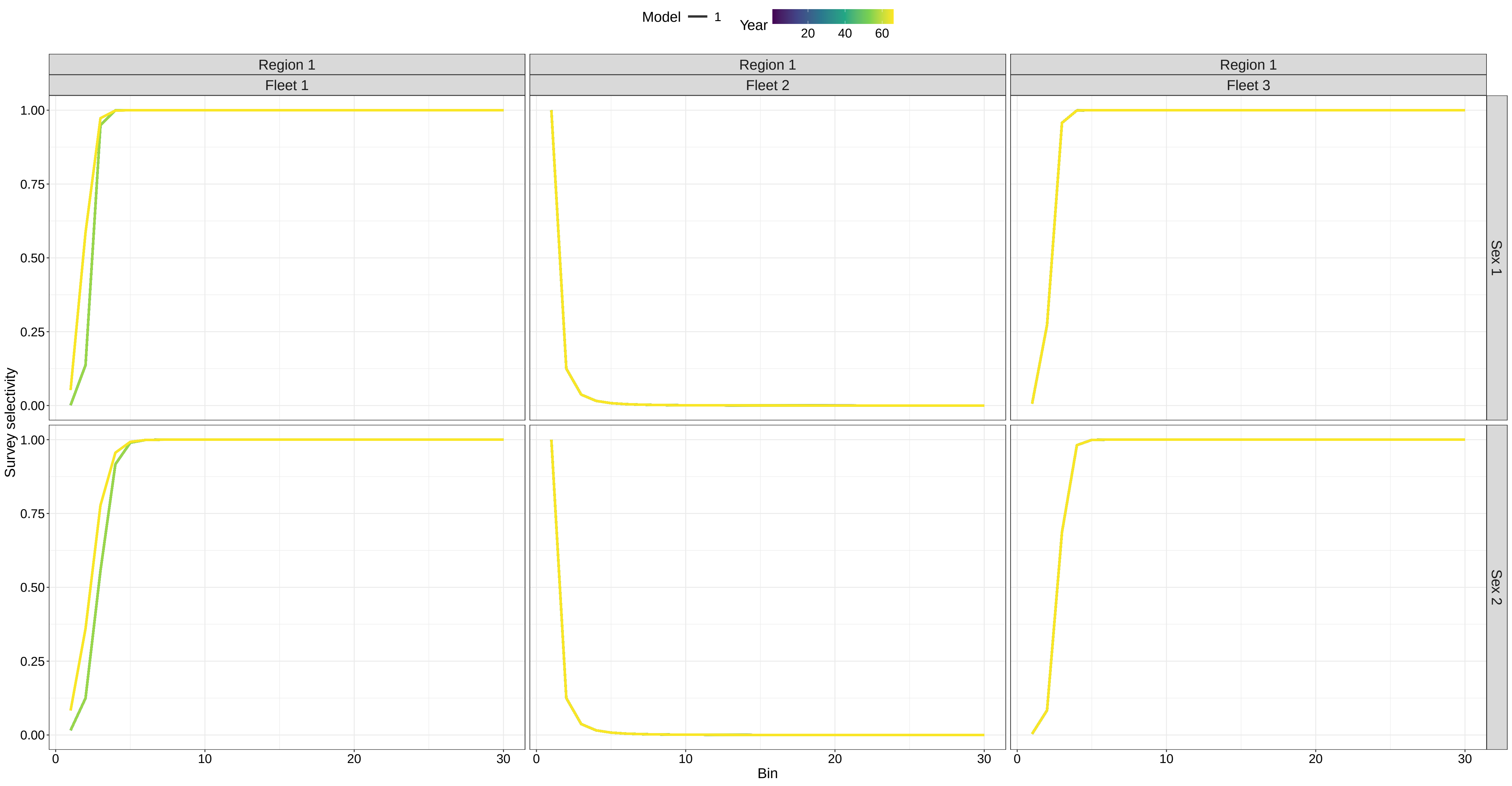
Numbers-at-Age

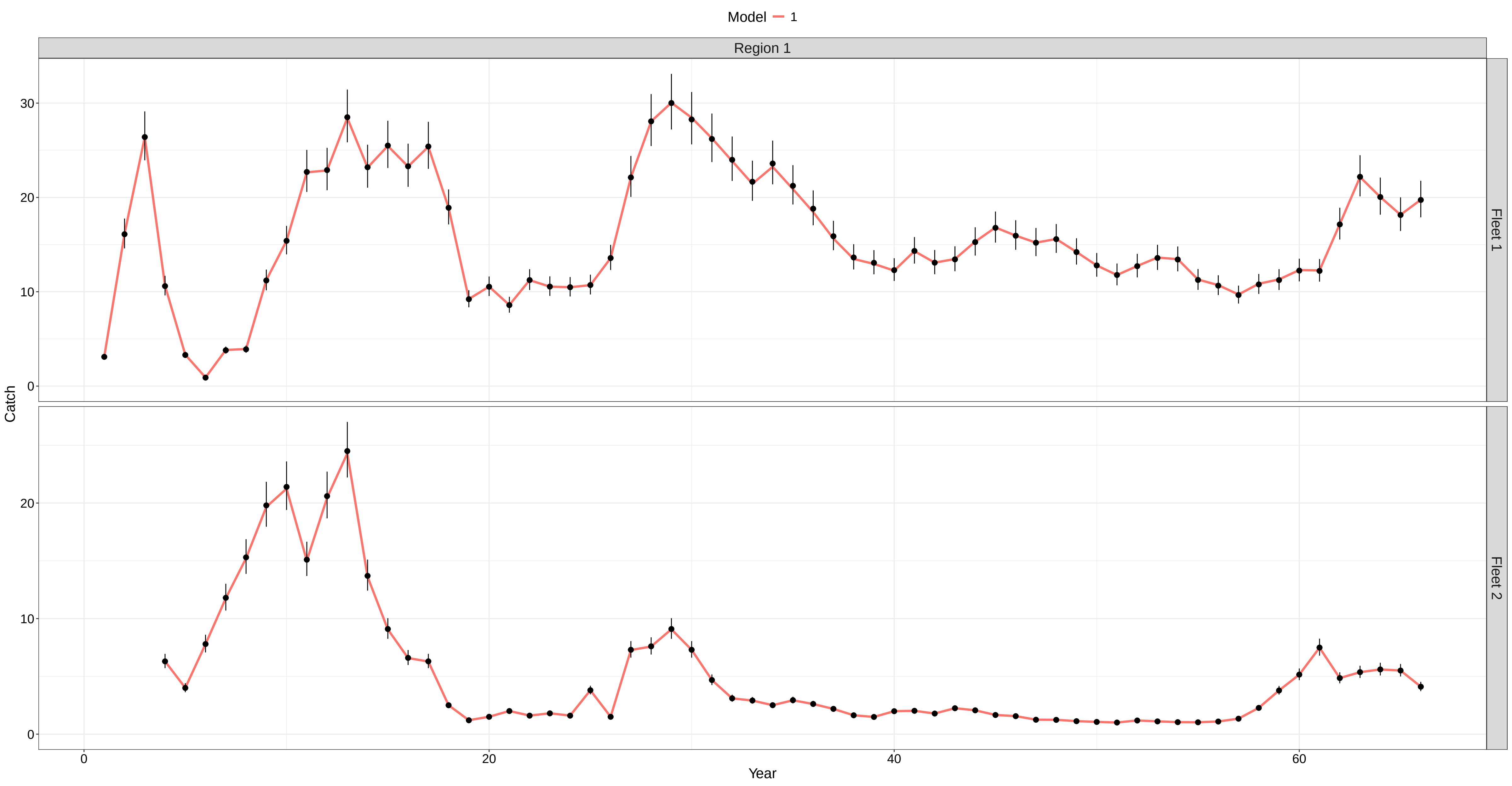
Sex — Female — Male

Numbers-at-age (millions)

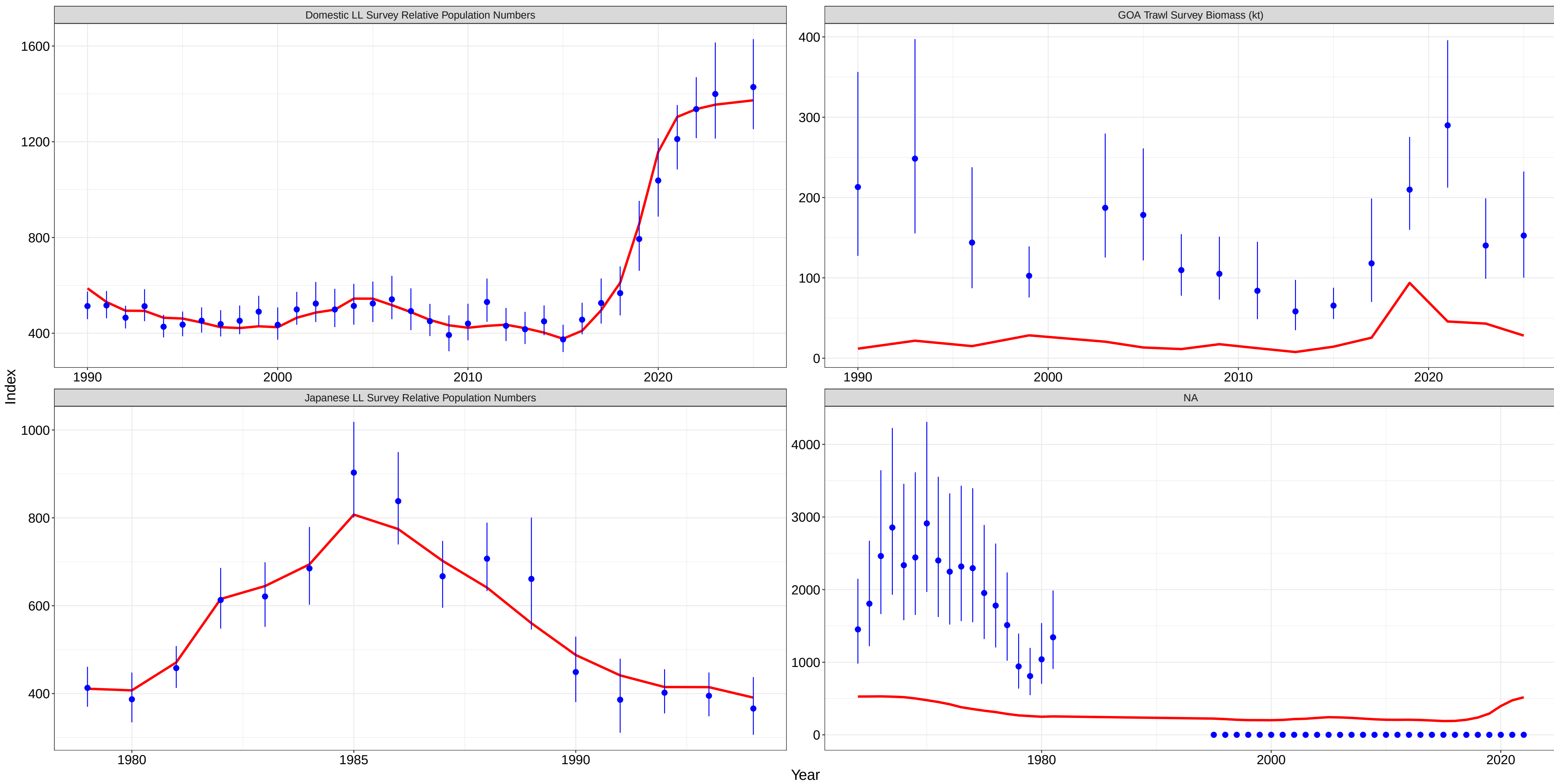




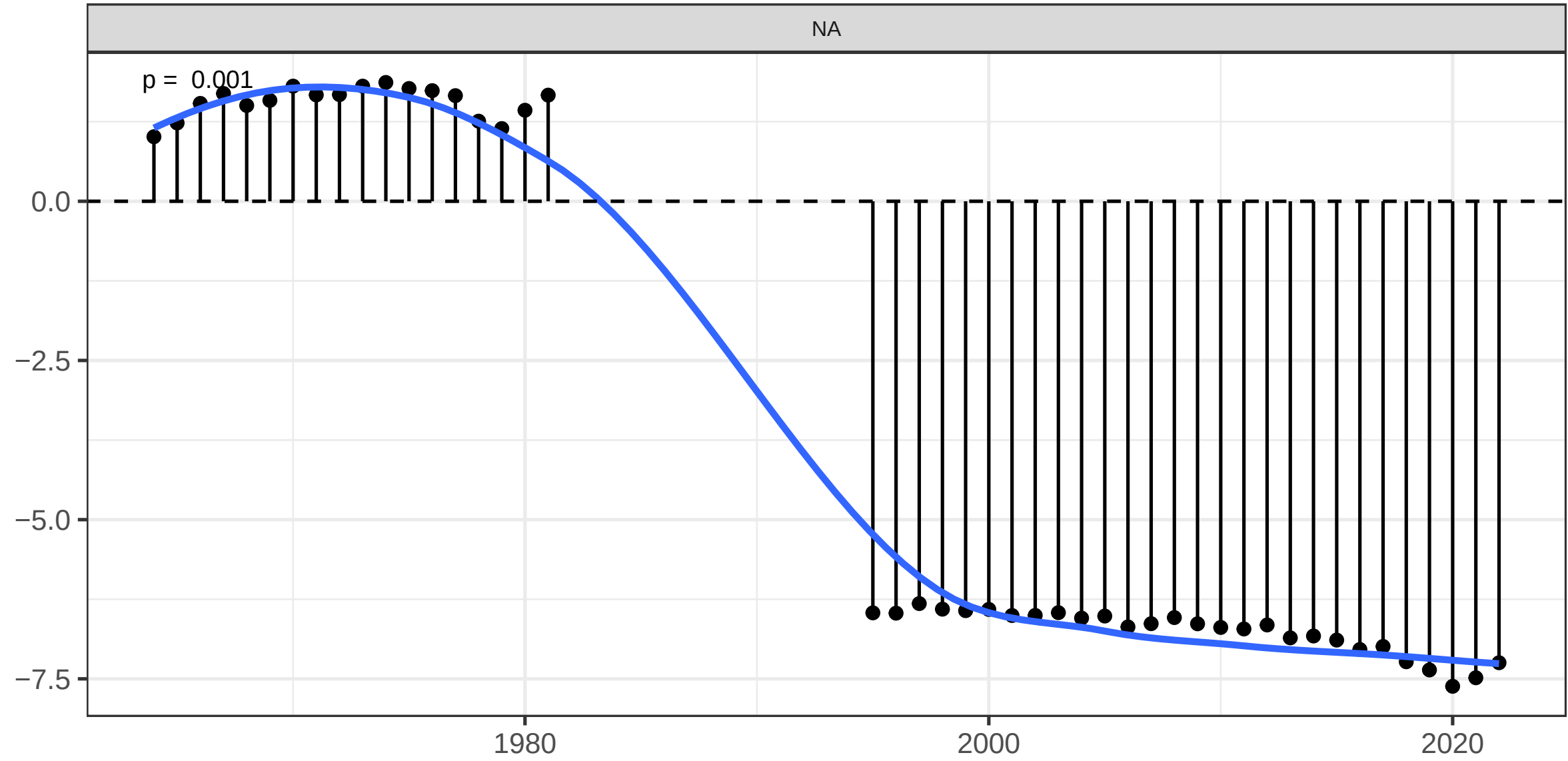
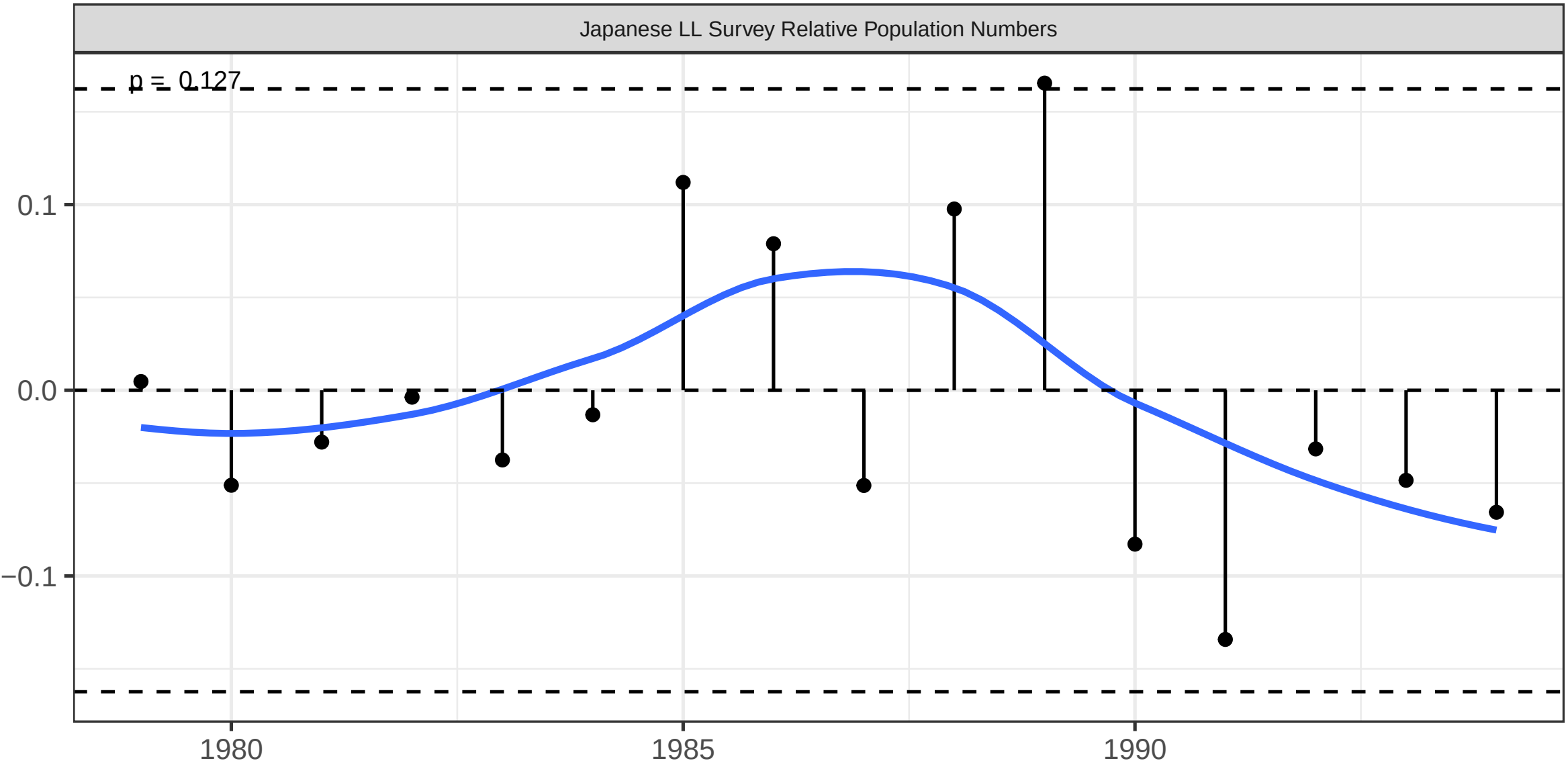
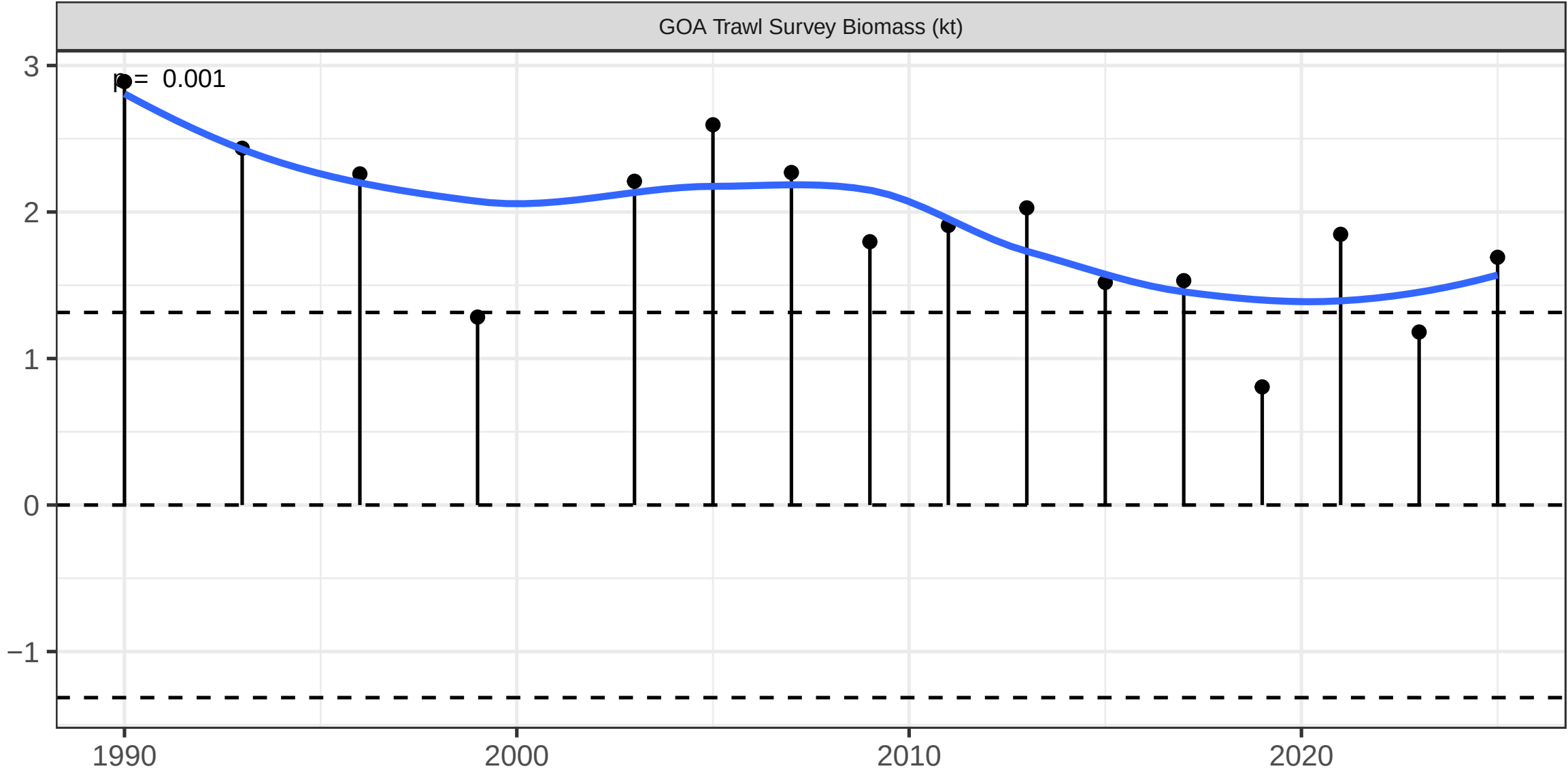
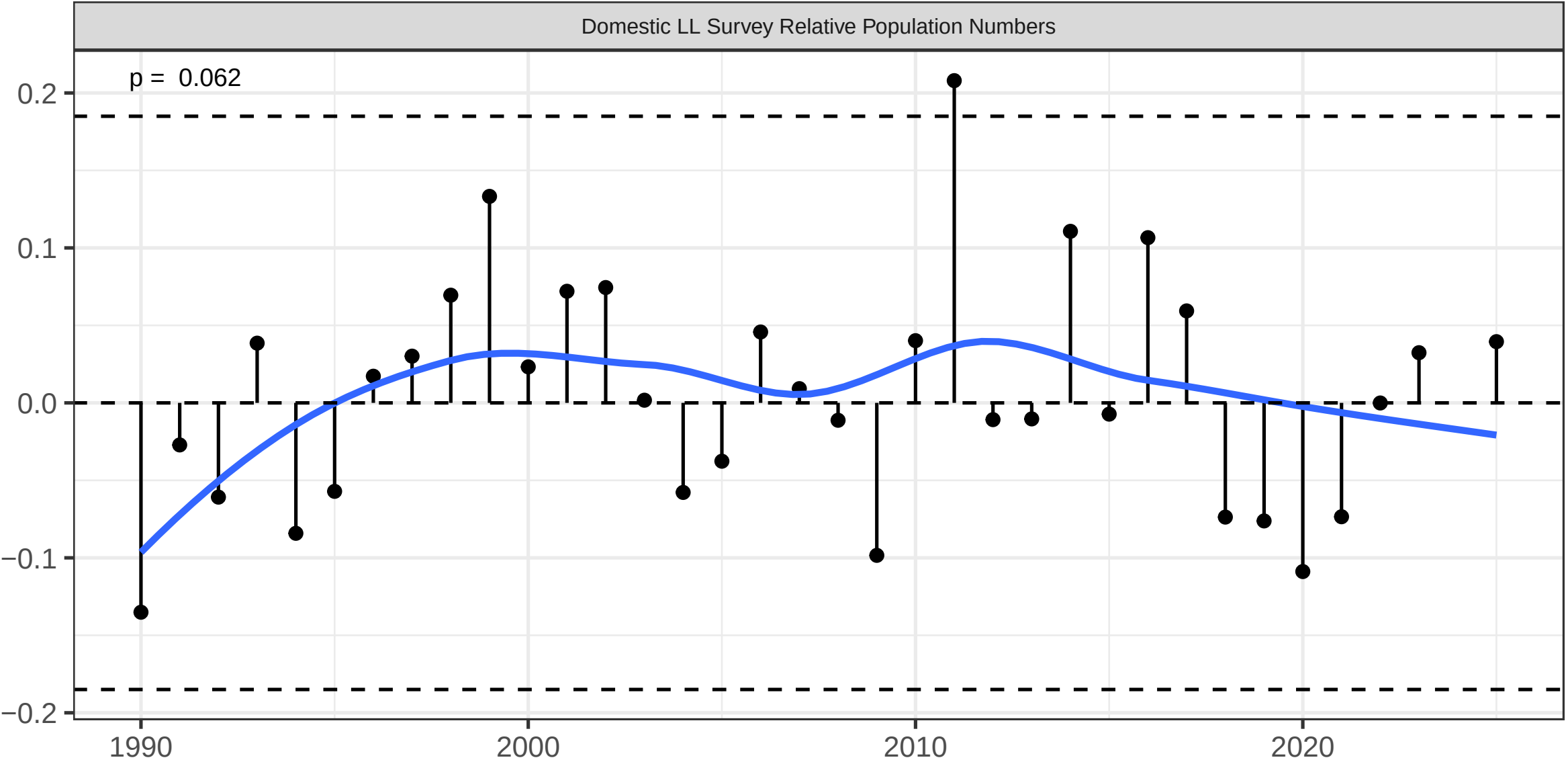




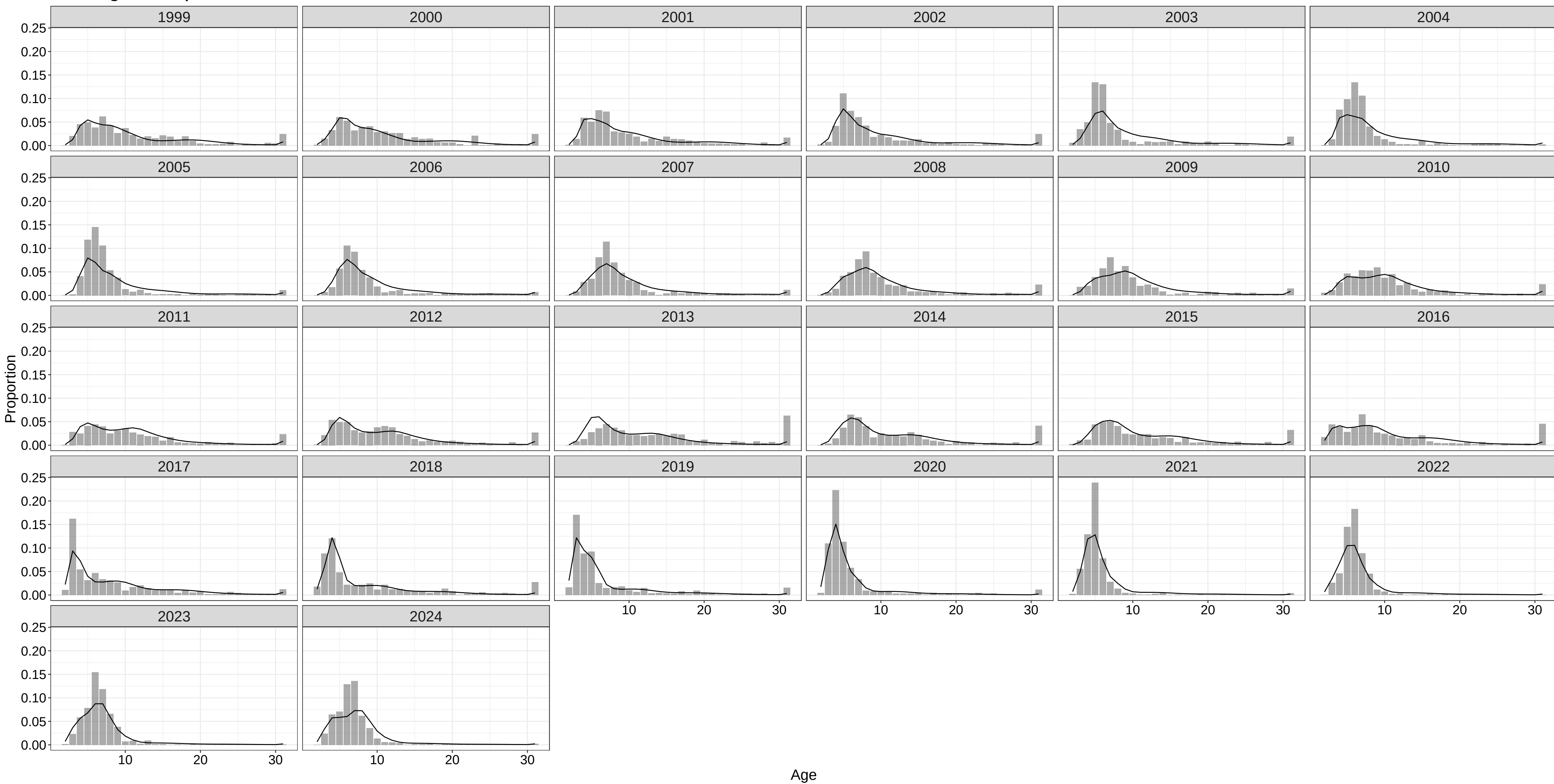
Index Fits



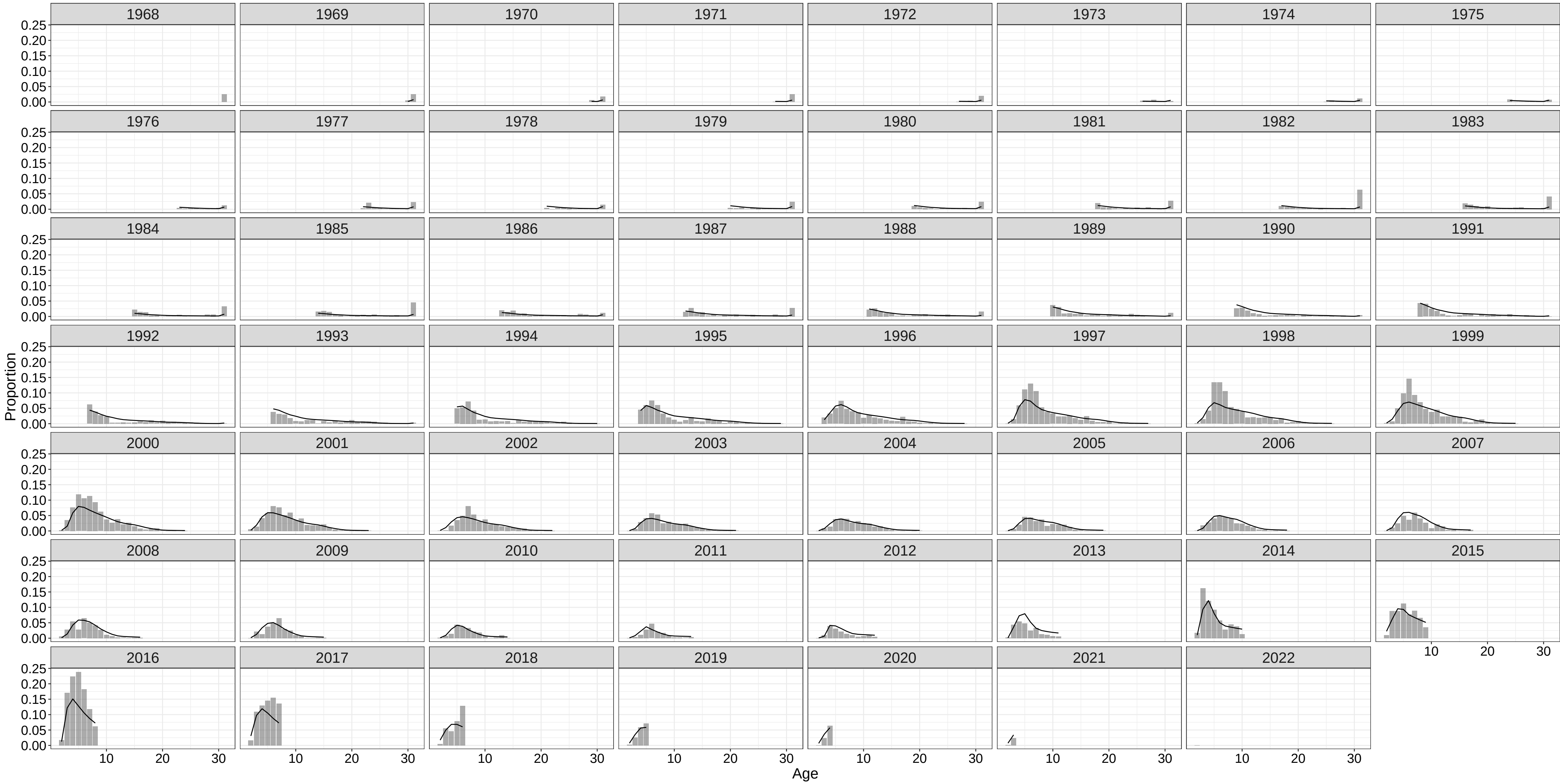
Index Residuals Runs Test



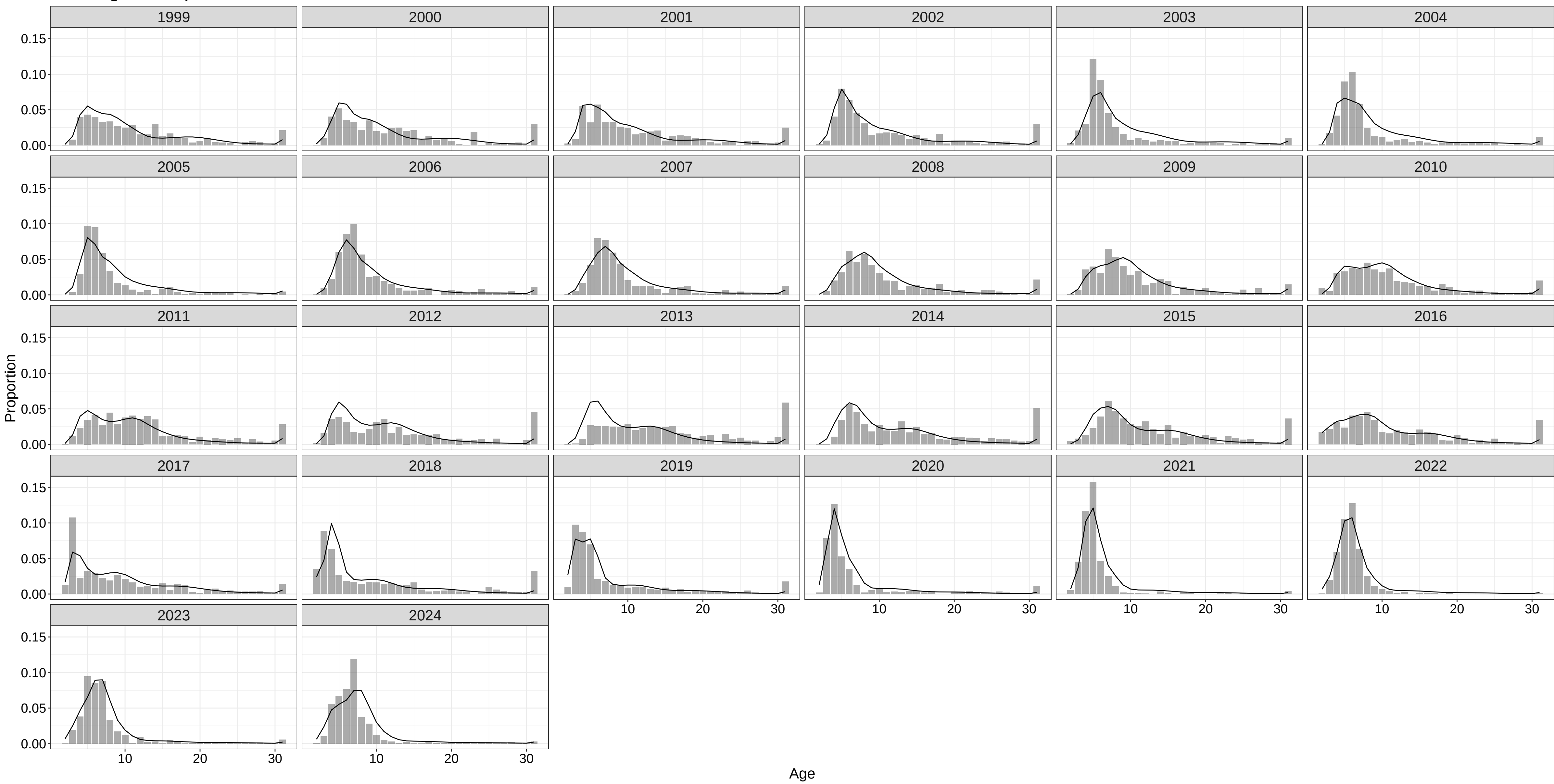
LLF Age Comps Female



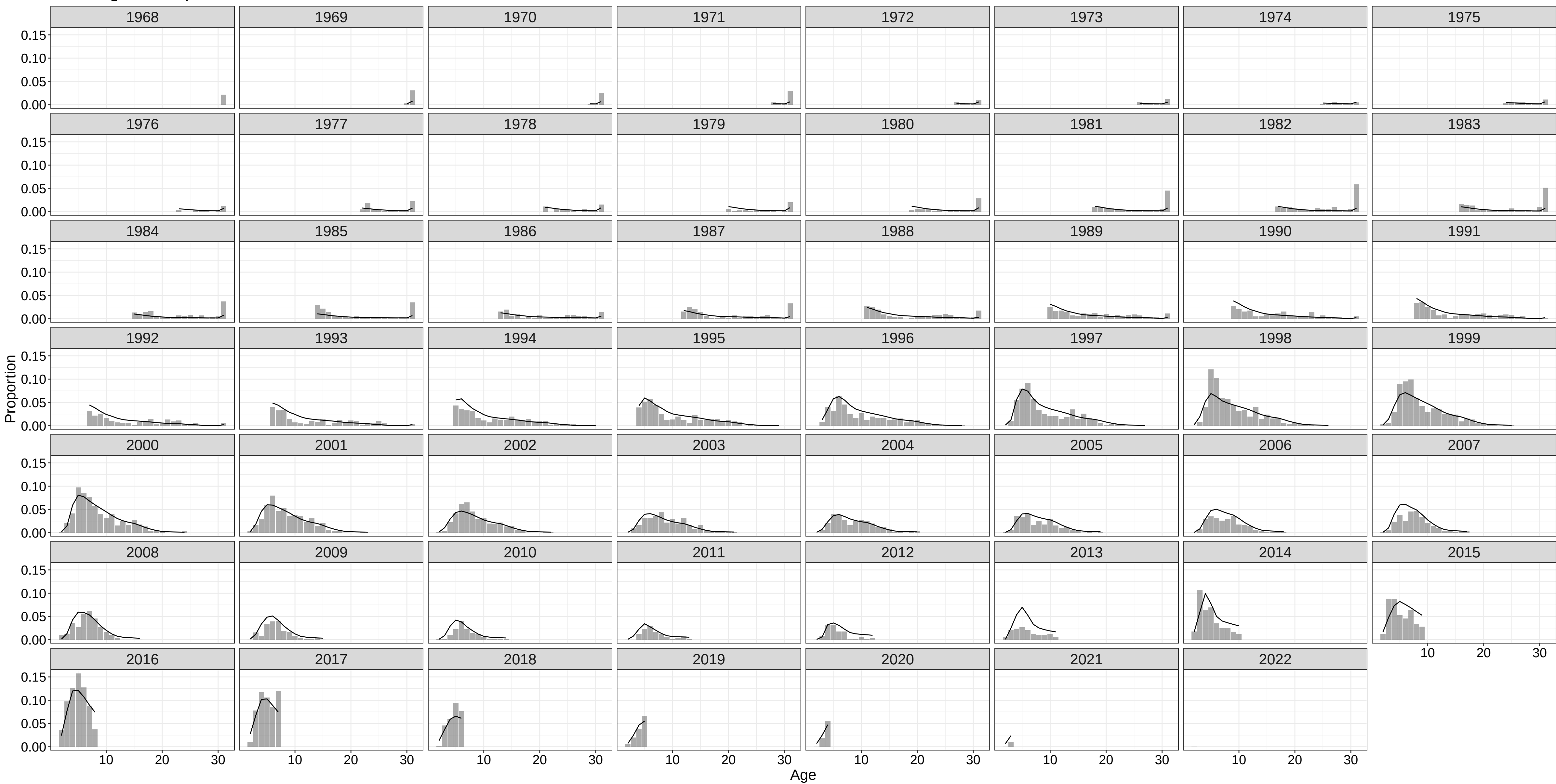
LLF Age Comps Cohort Fits Female



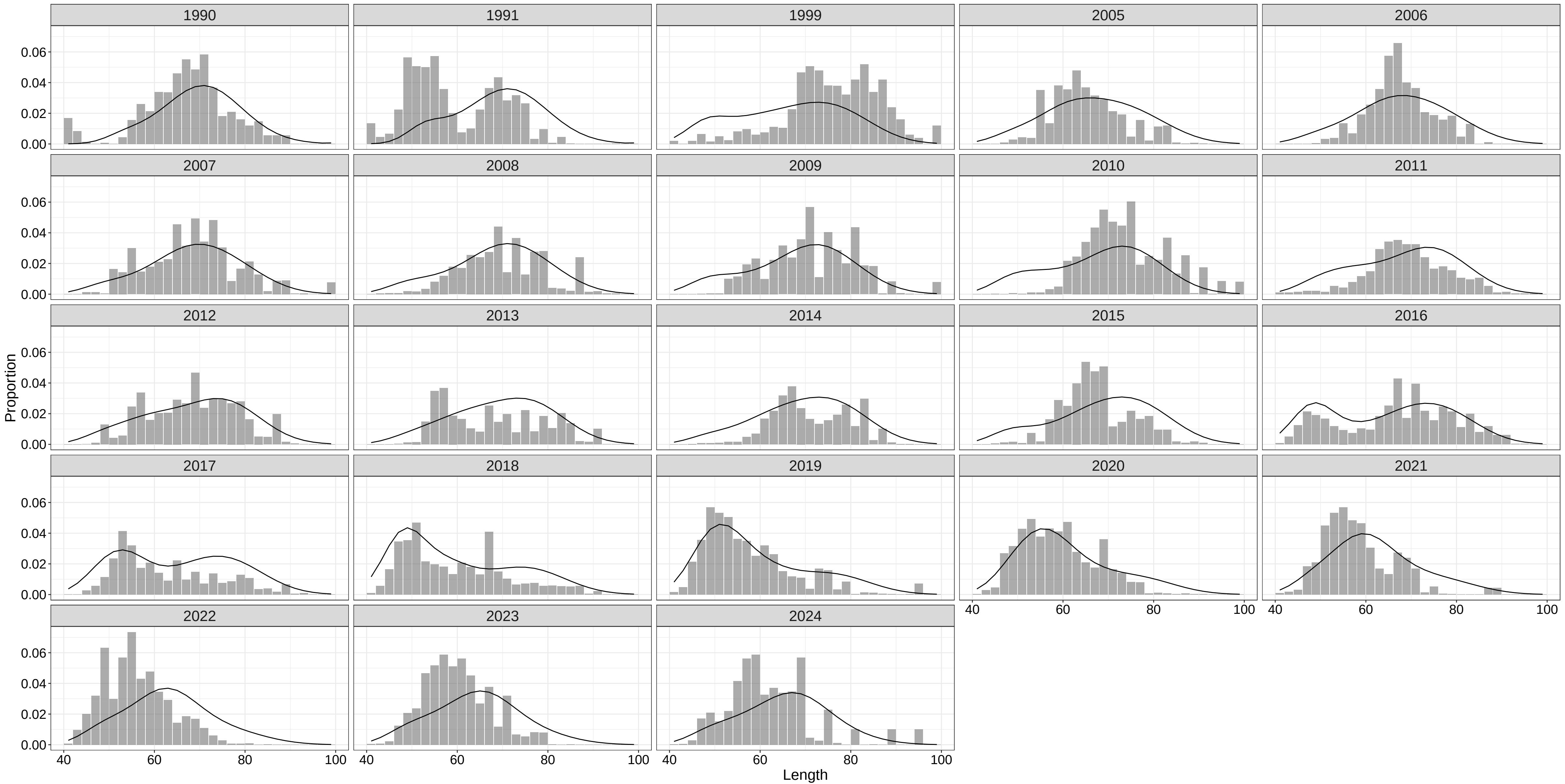
LLF Age Comps Male



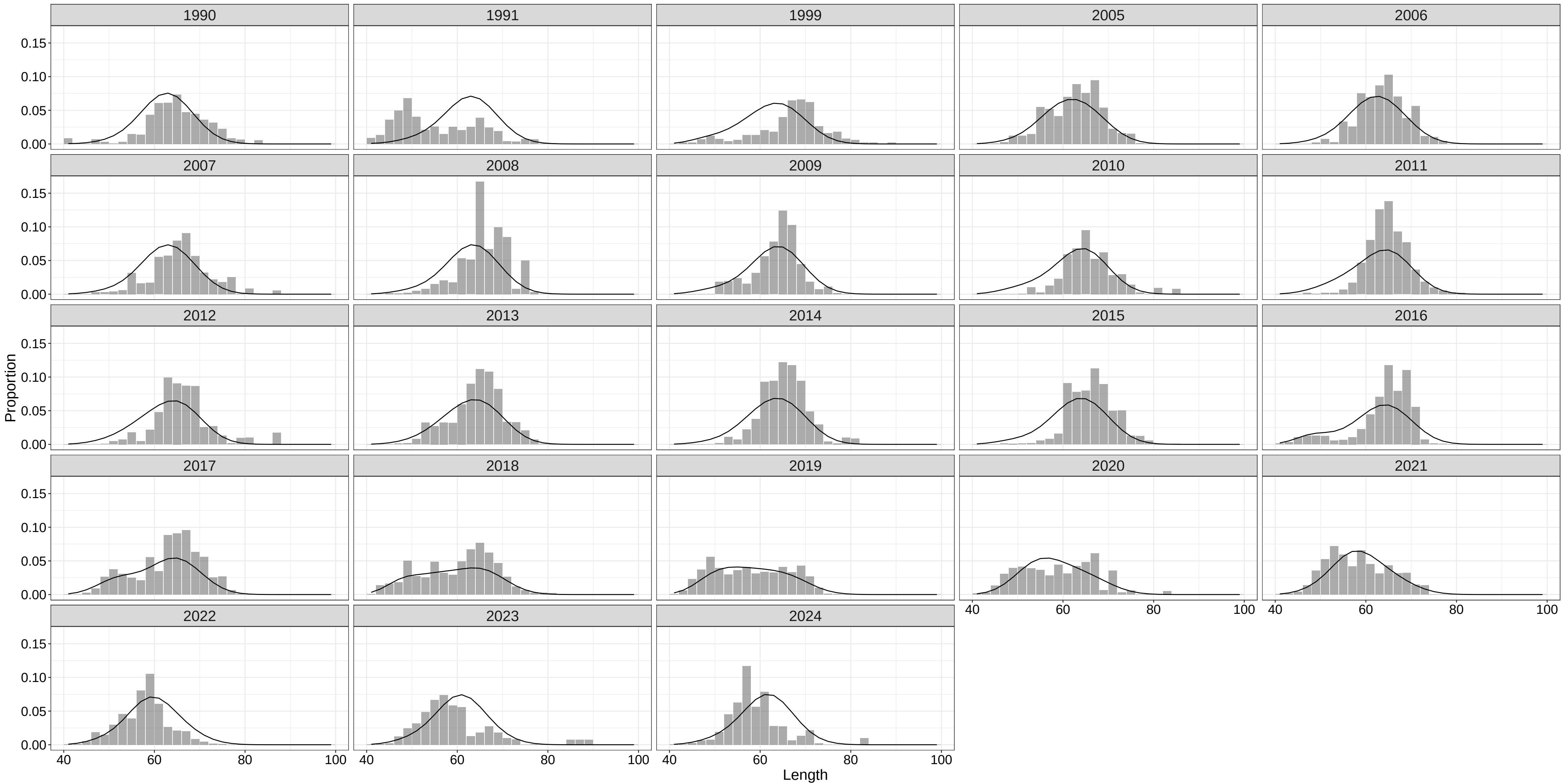
LLF Age Comps Cohort Fits Male



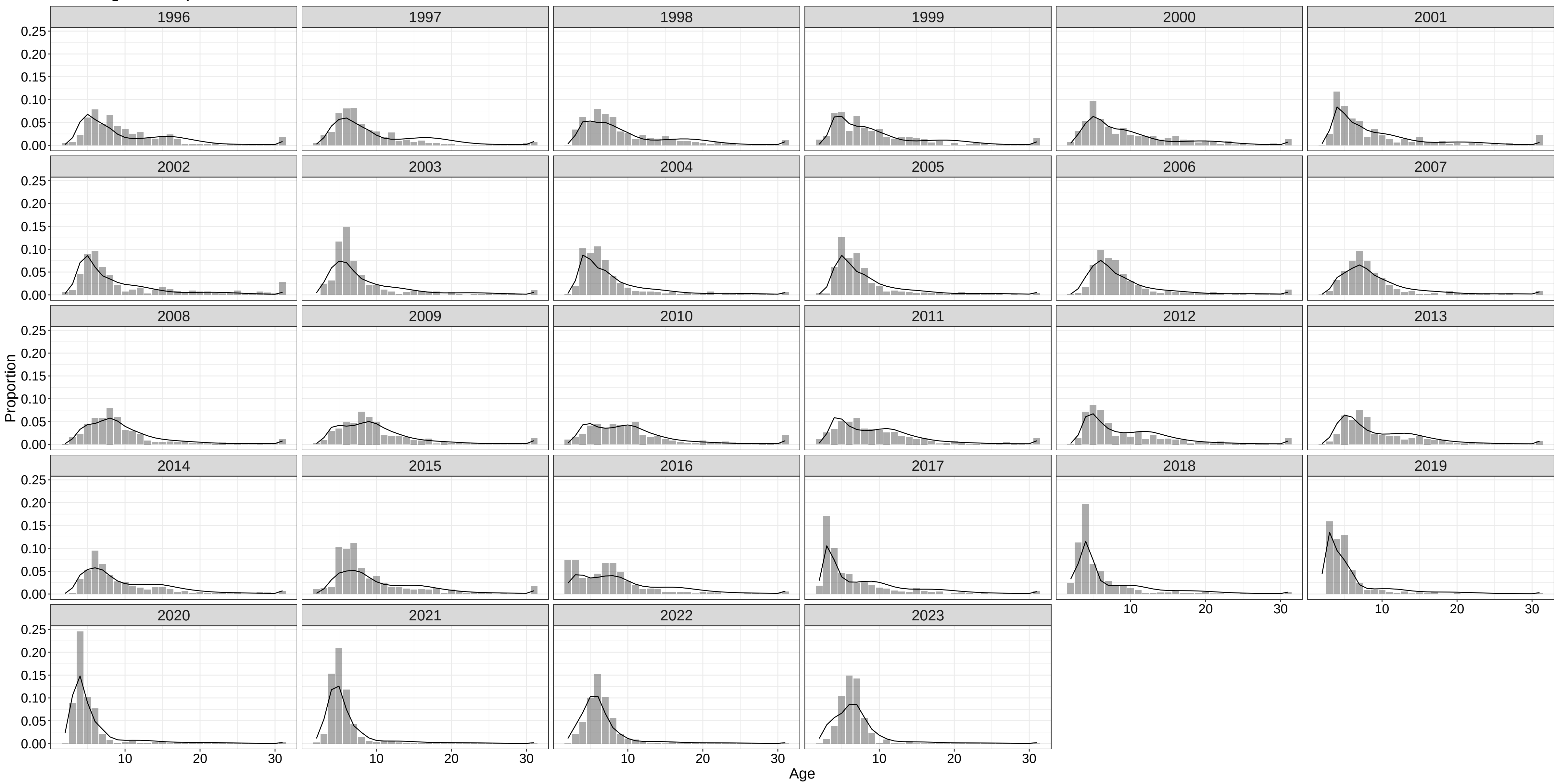
TF Length Comps Female



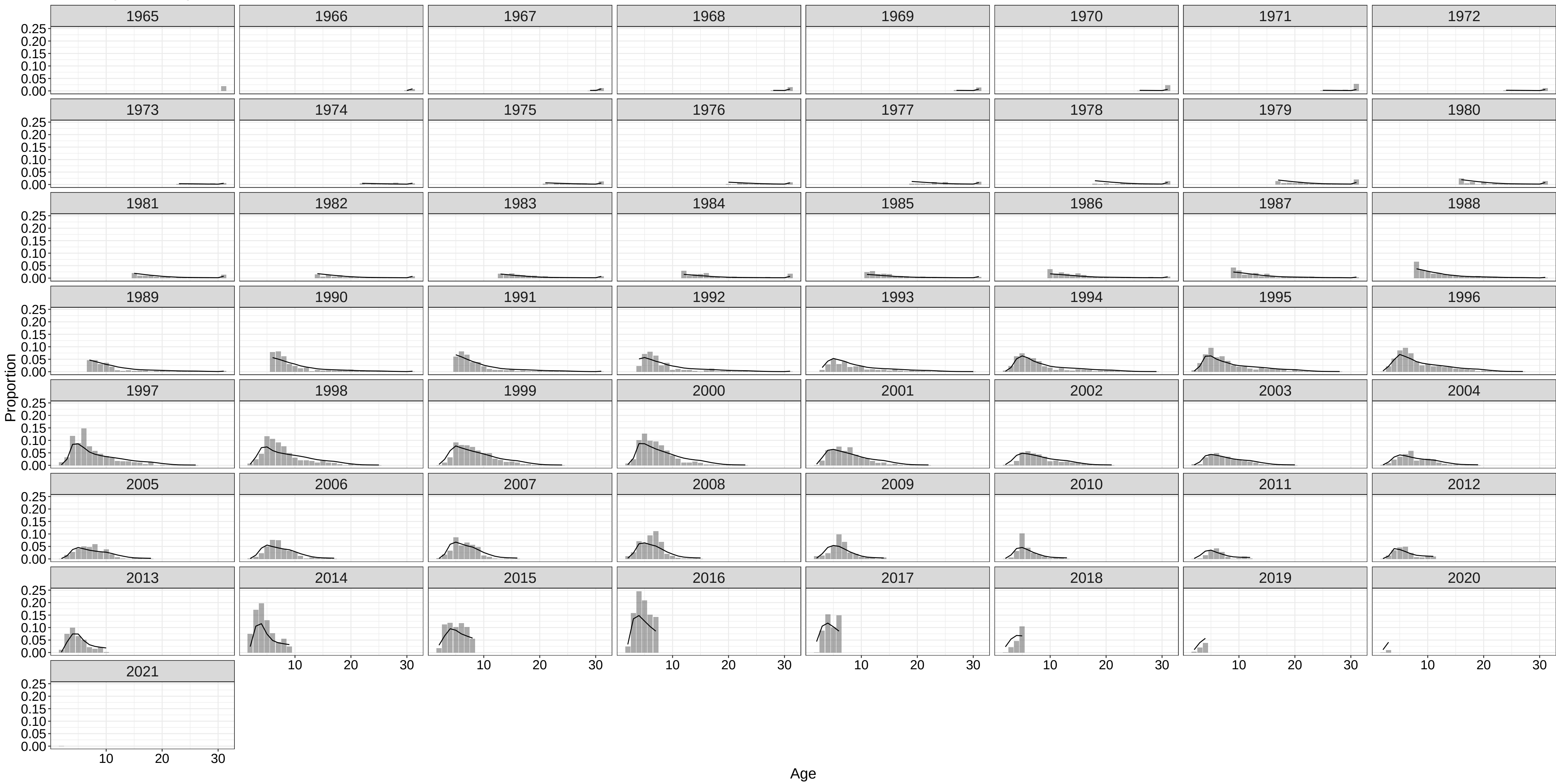
TF Length Comps Male



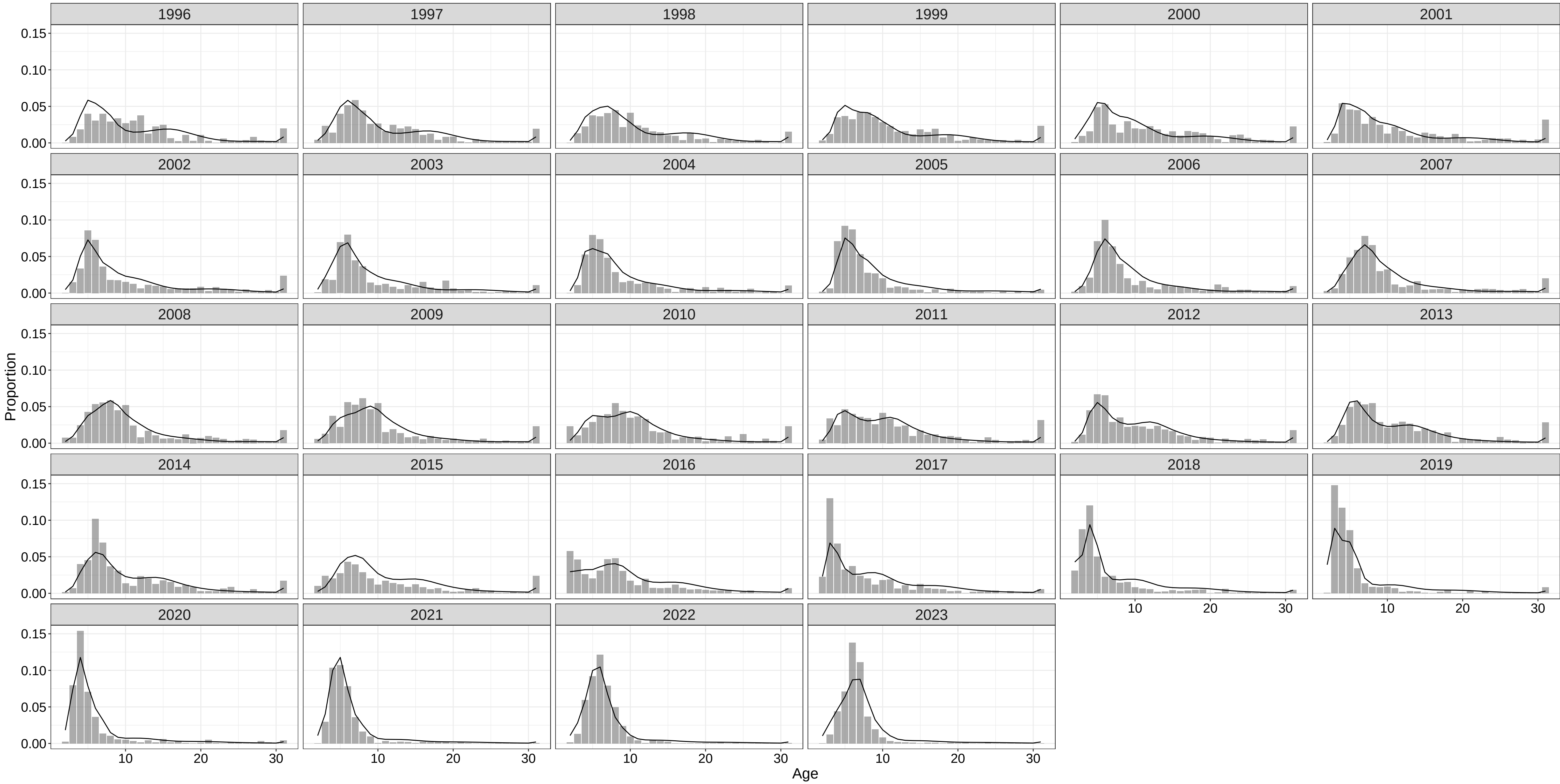
LLS Age Comps Female



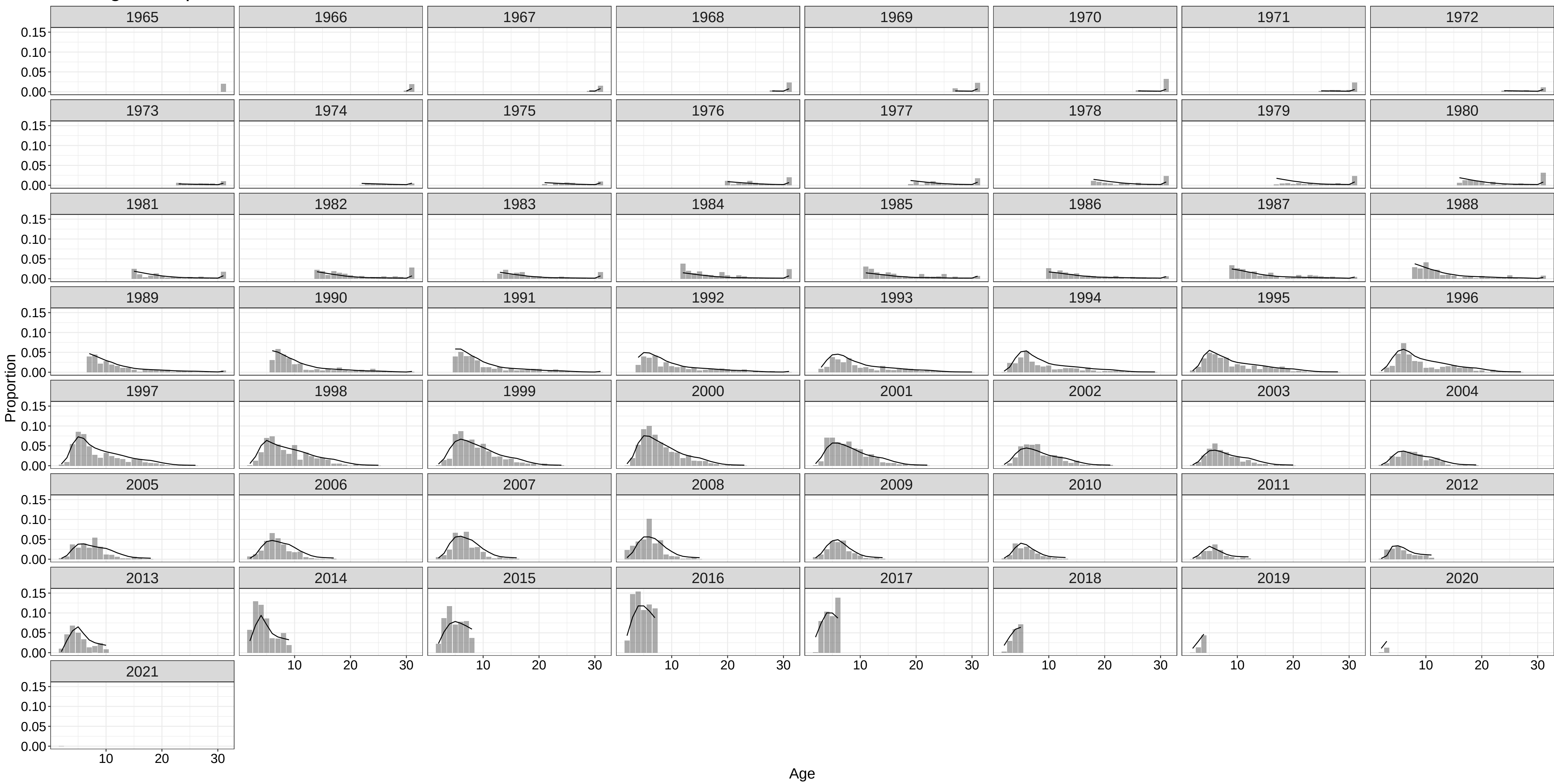
LLS Age Comps Cohort Fit Female



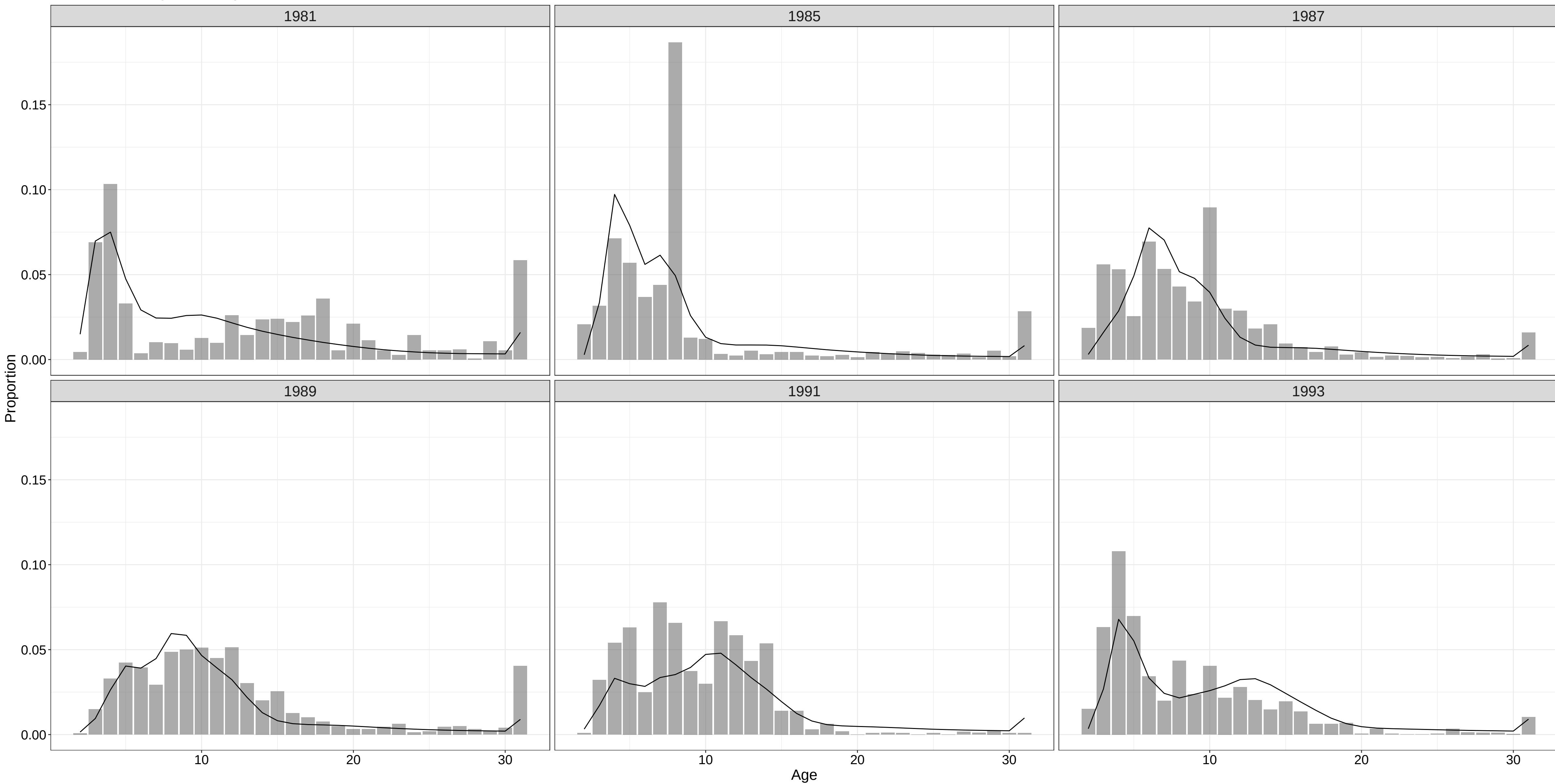
LLS Age Comps Male



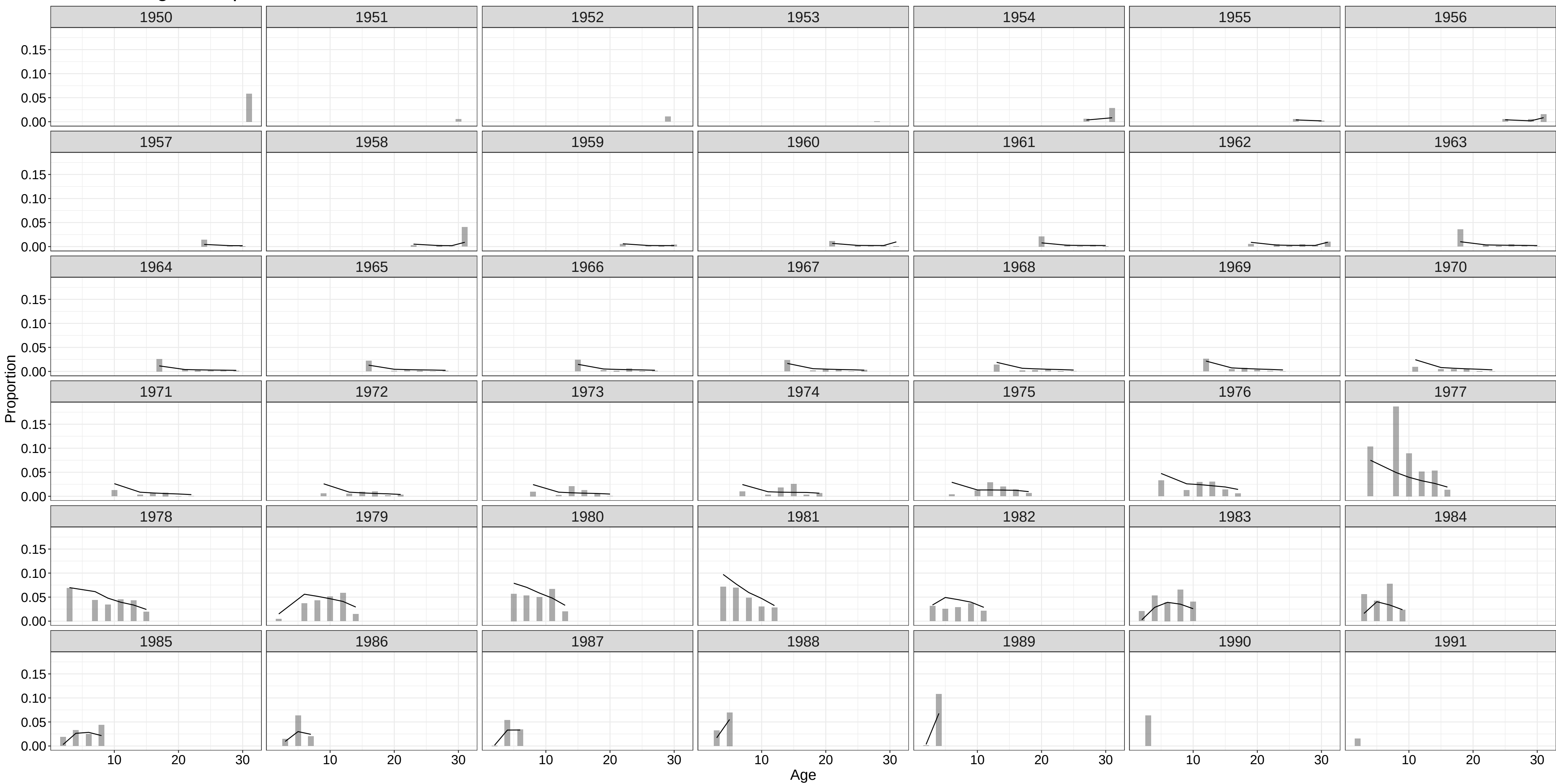
LLS Age Comps Cohort Fit Male



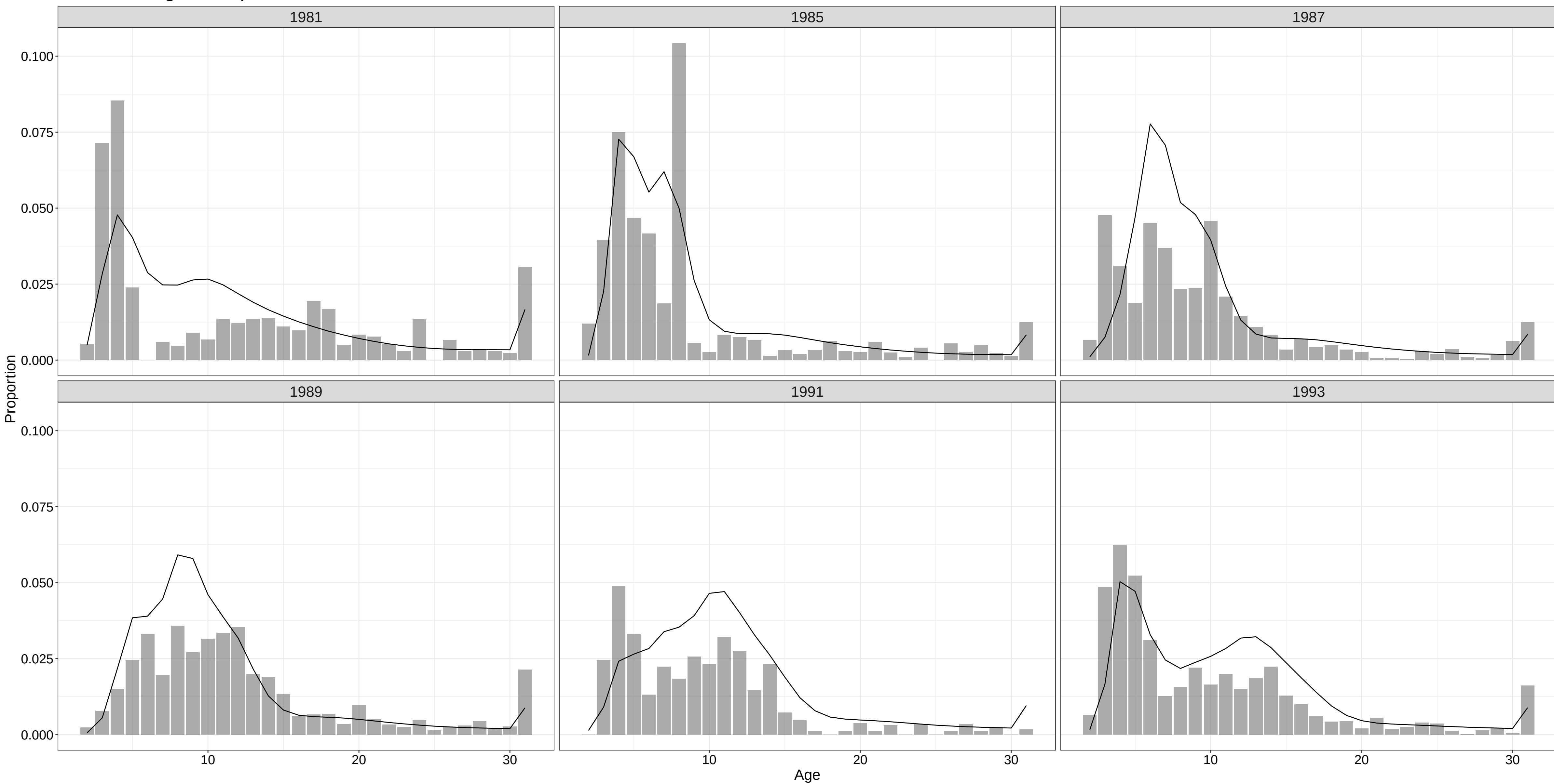
JPN LLS Age Comps Female



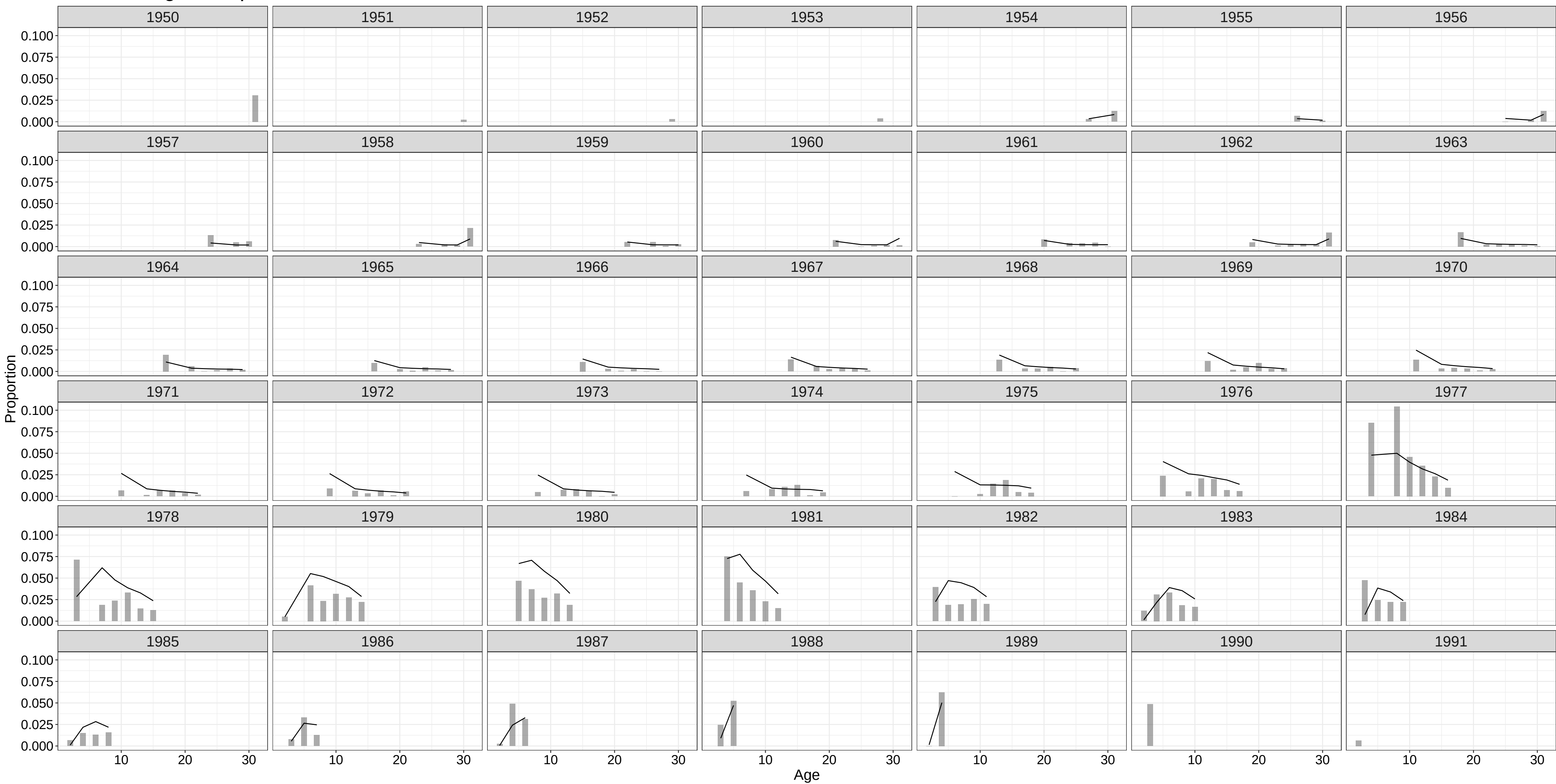
JPN LLS Age Comps Cohort Fit Female



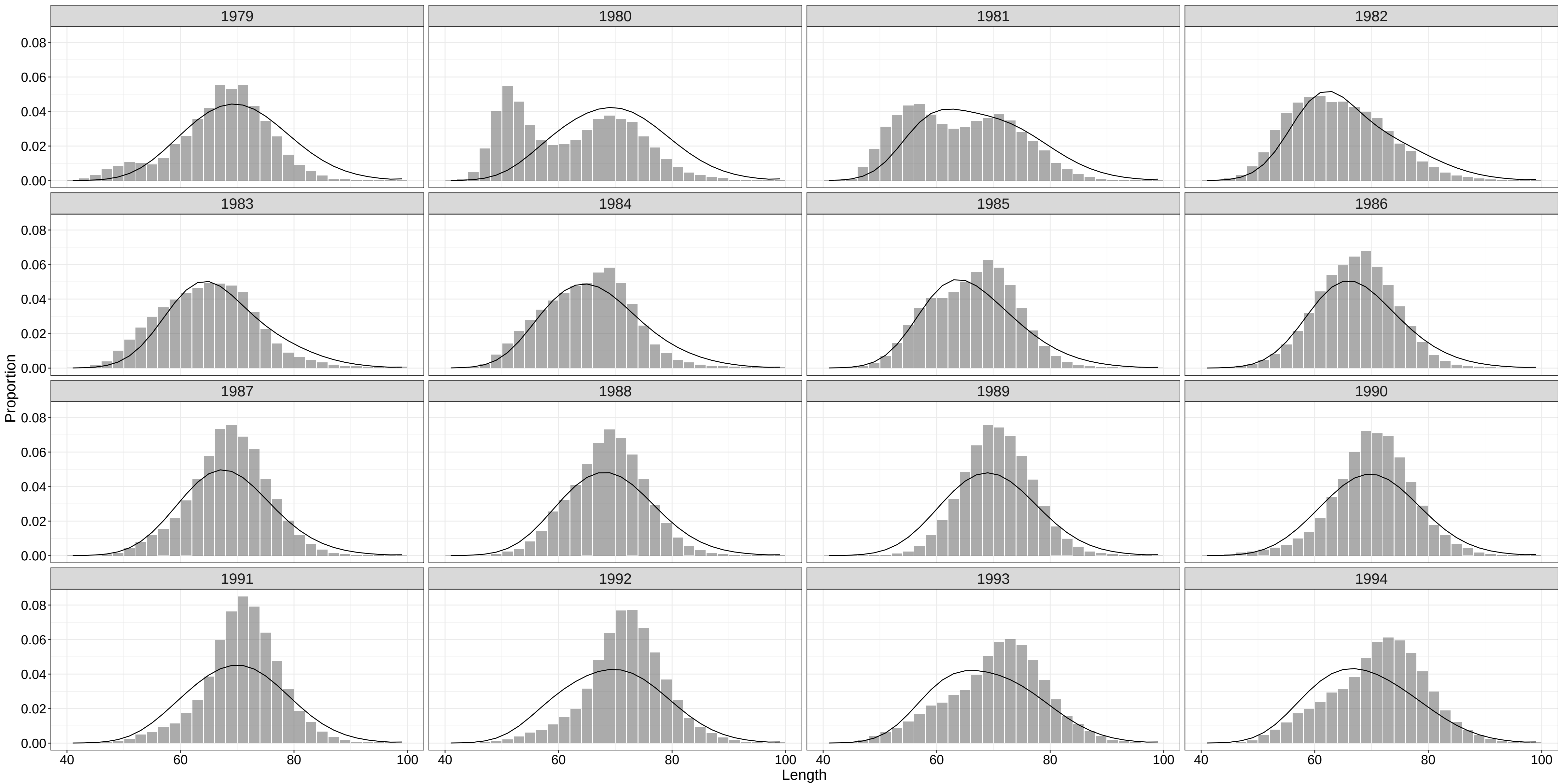
JPN LLS Age Comps Male



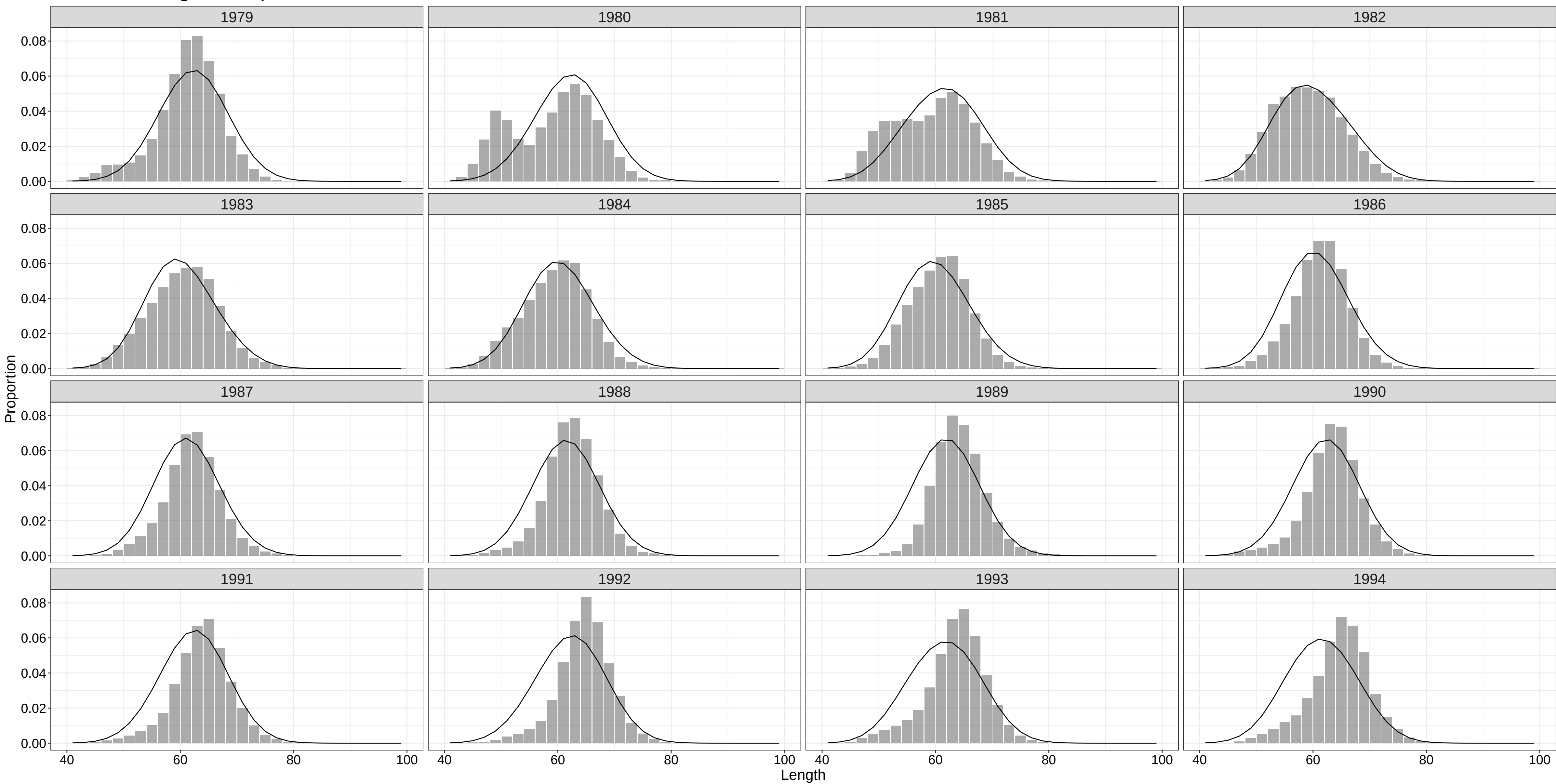
JPN LLS Age Comps Cohort Fit Male



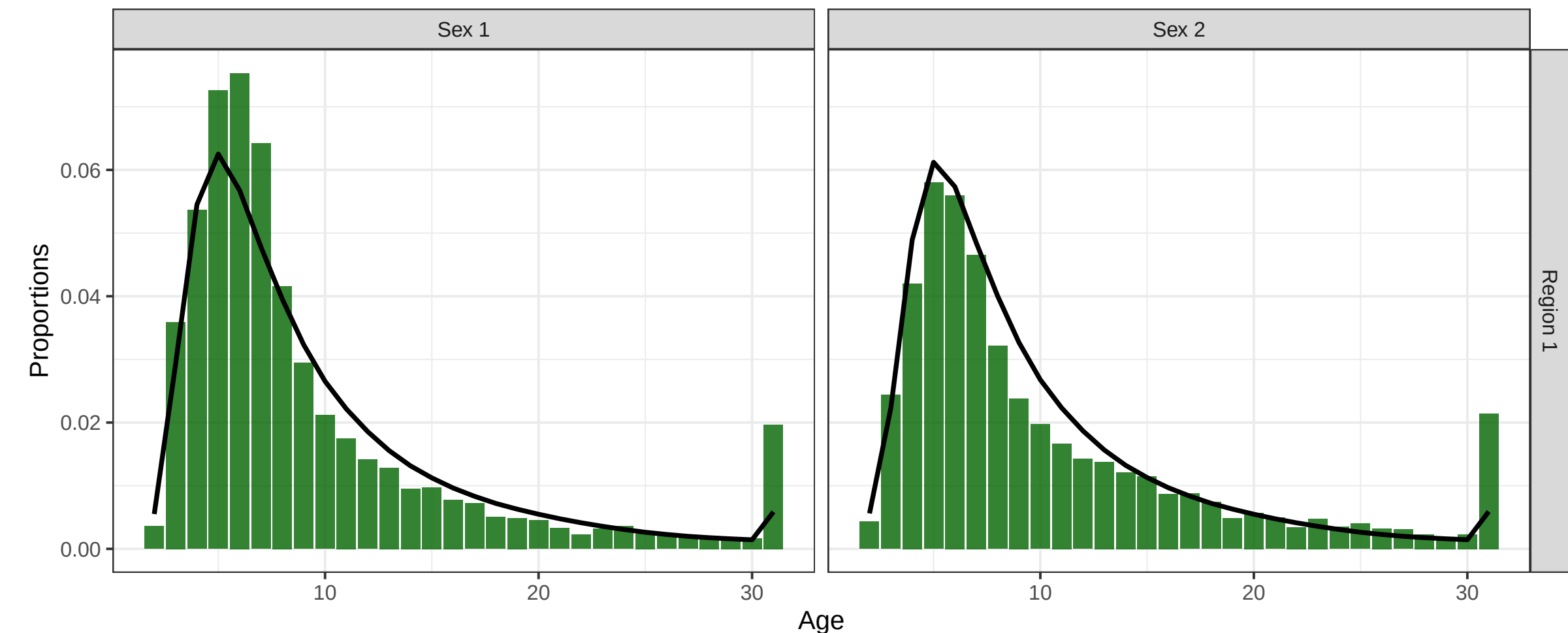
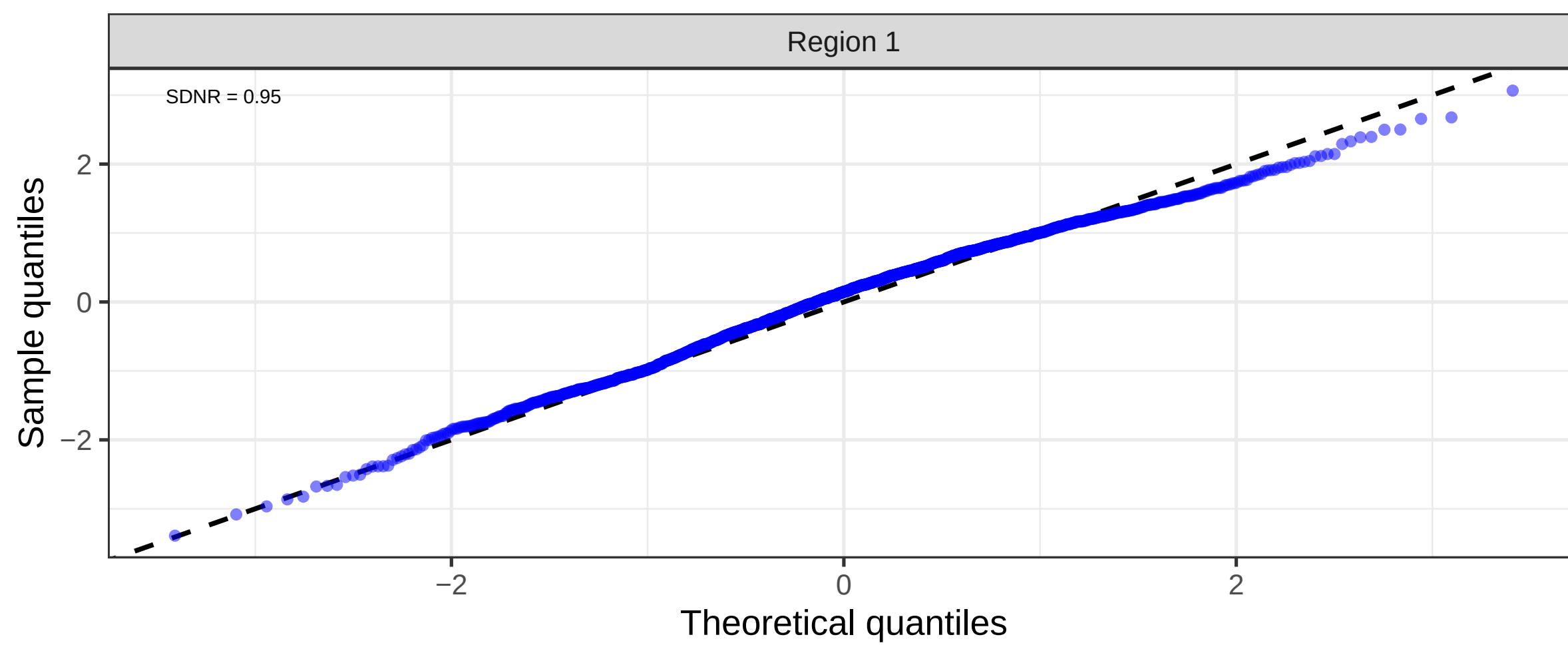
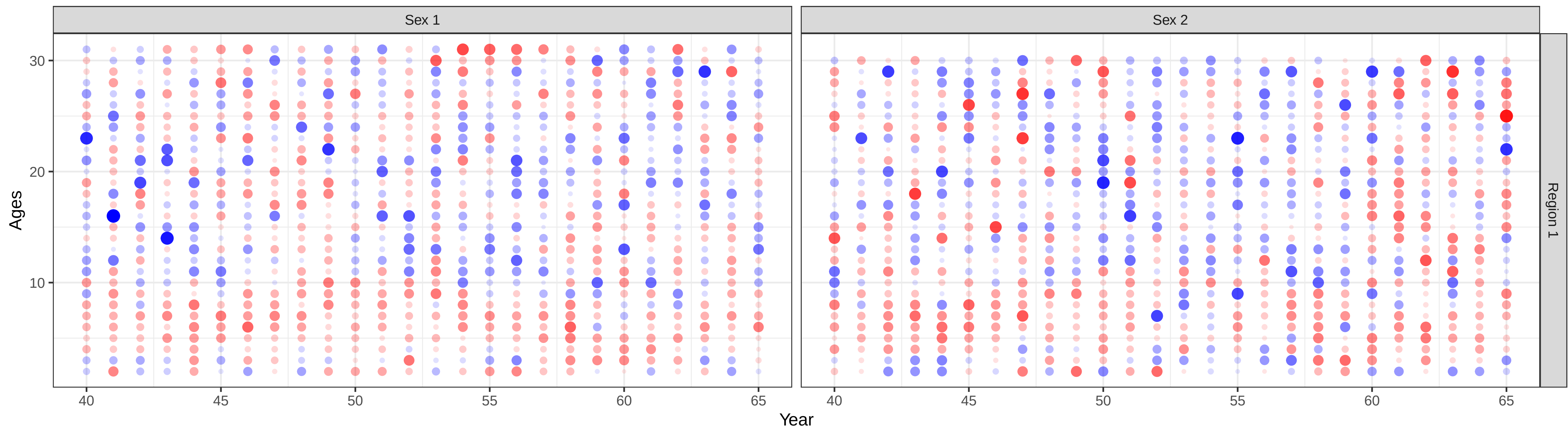
JPN LLS Length Comps Female

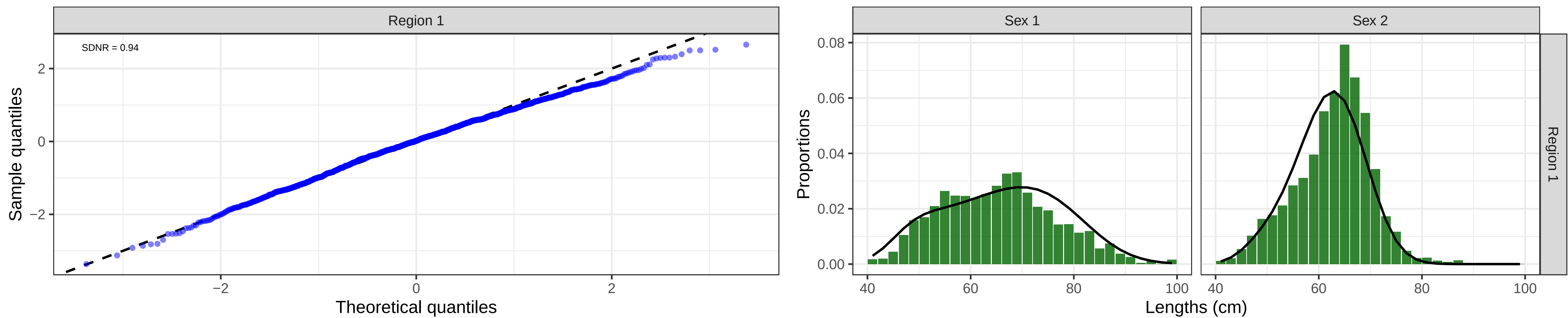
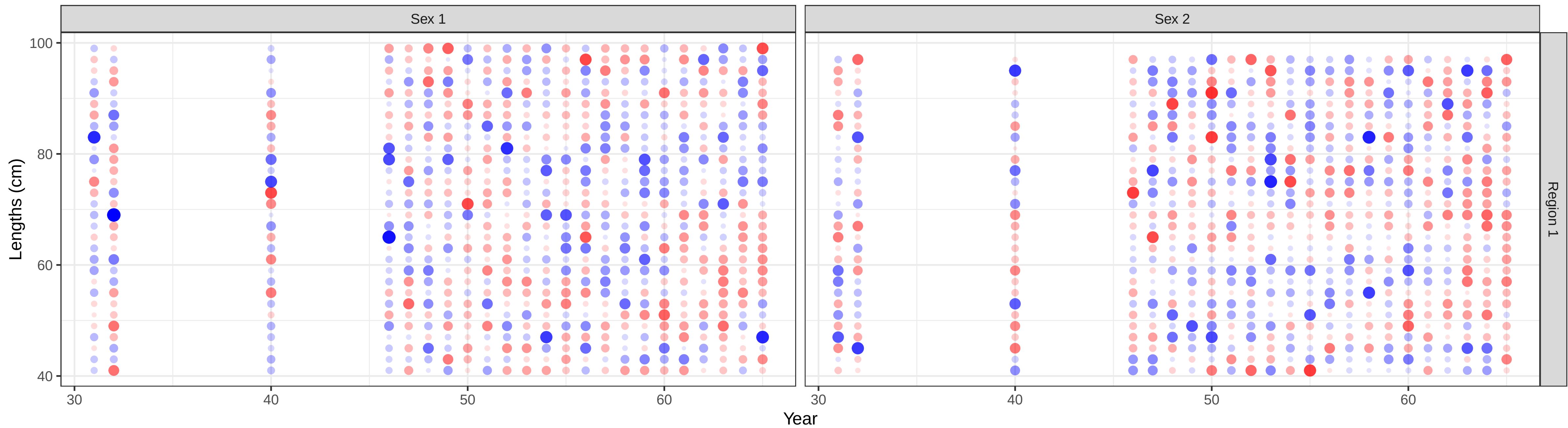


JPN LLS Length Comps Male

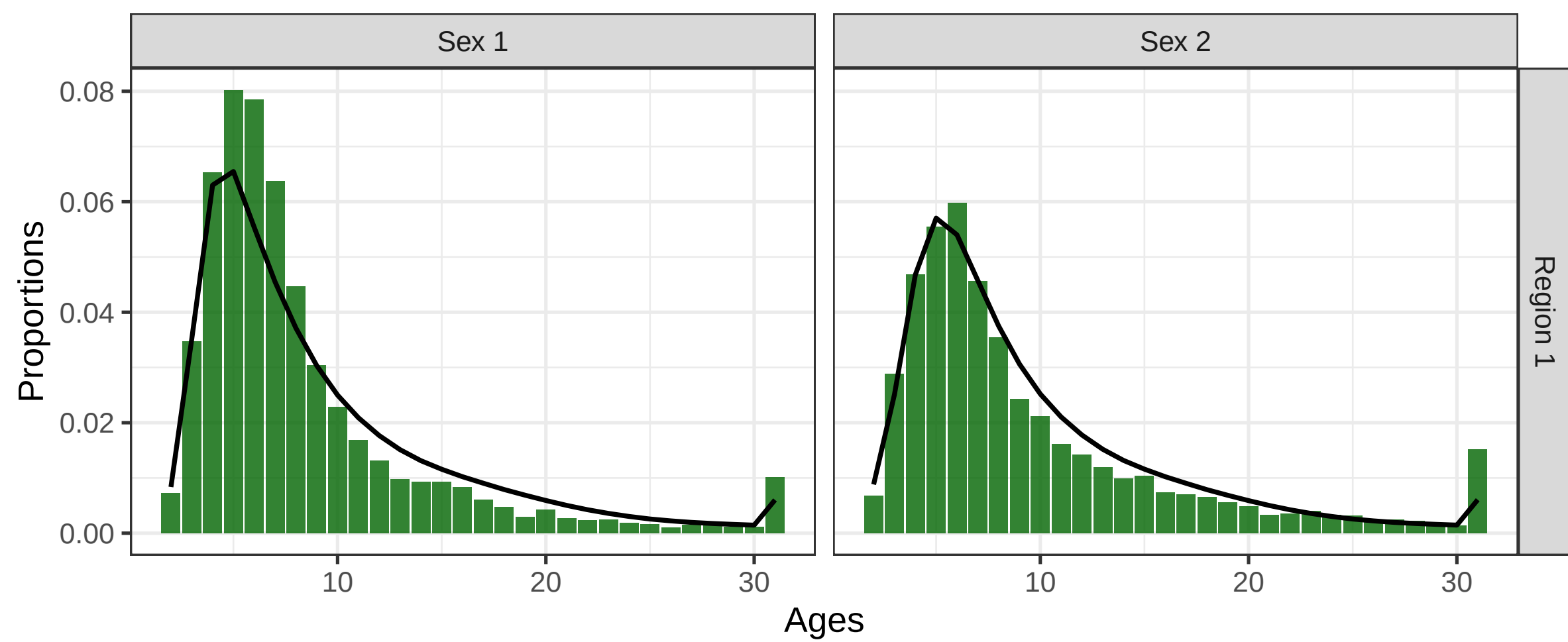
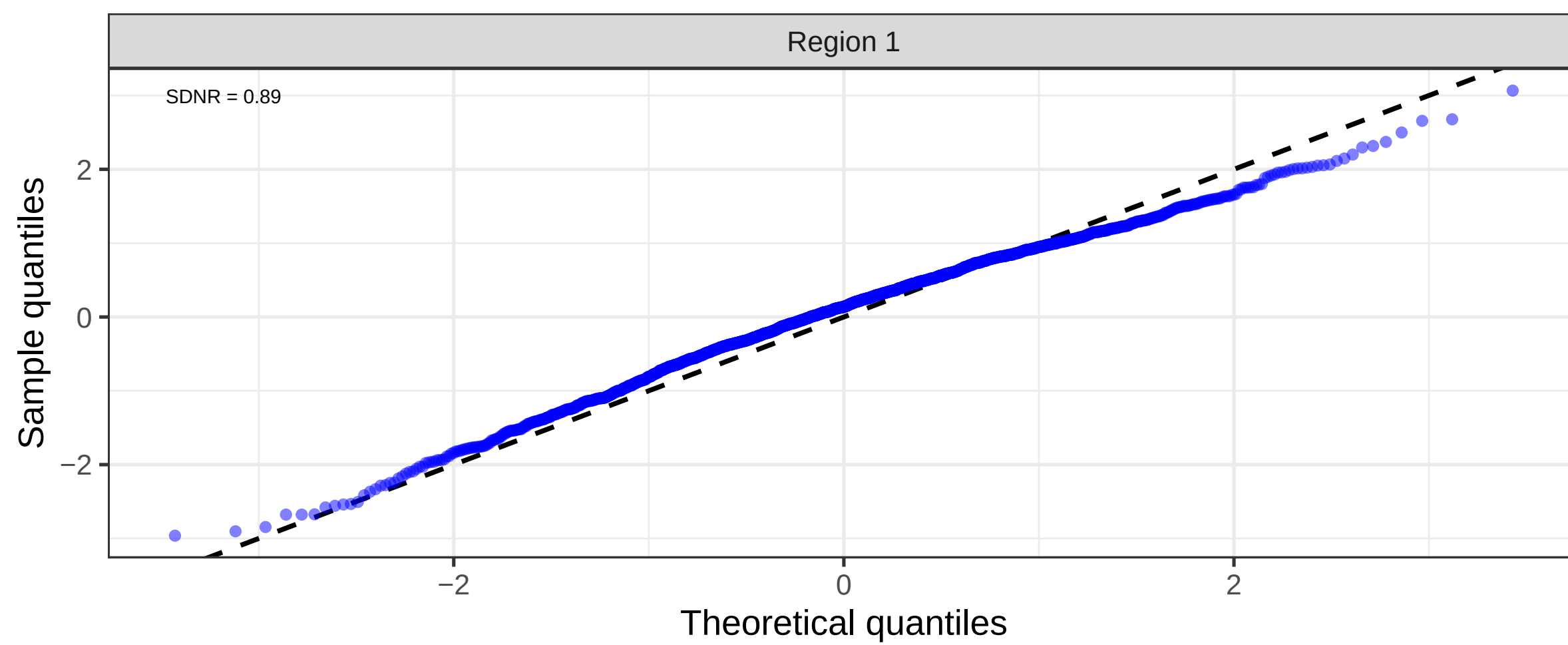
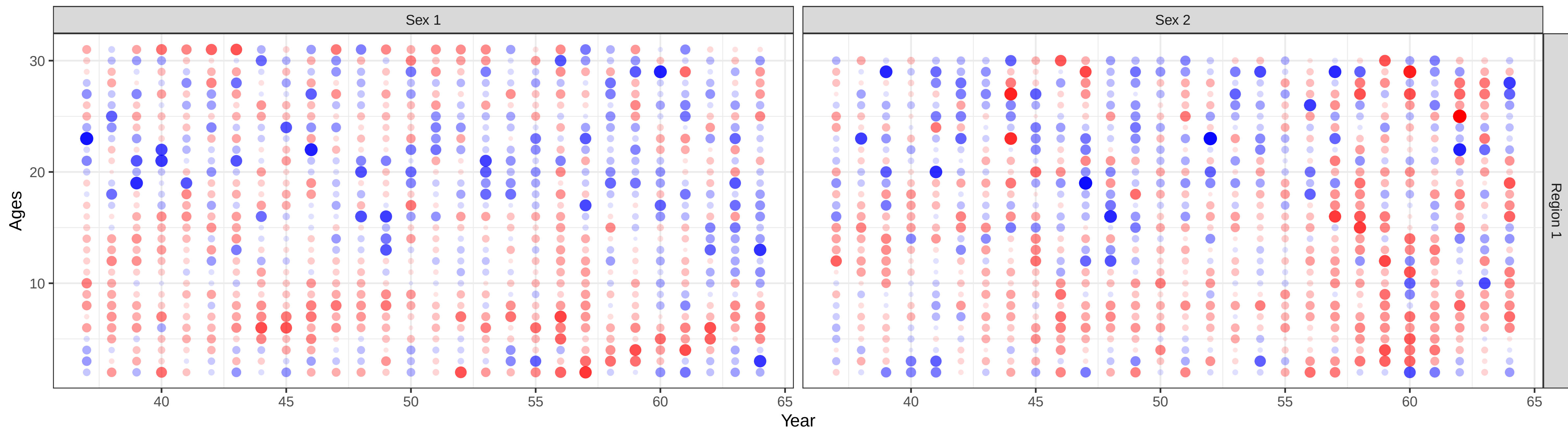


abs(Resid) 1 2 3 Sign ● Neg ● Pos

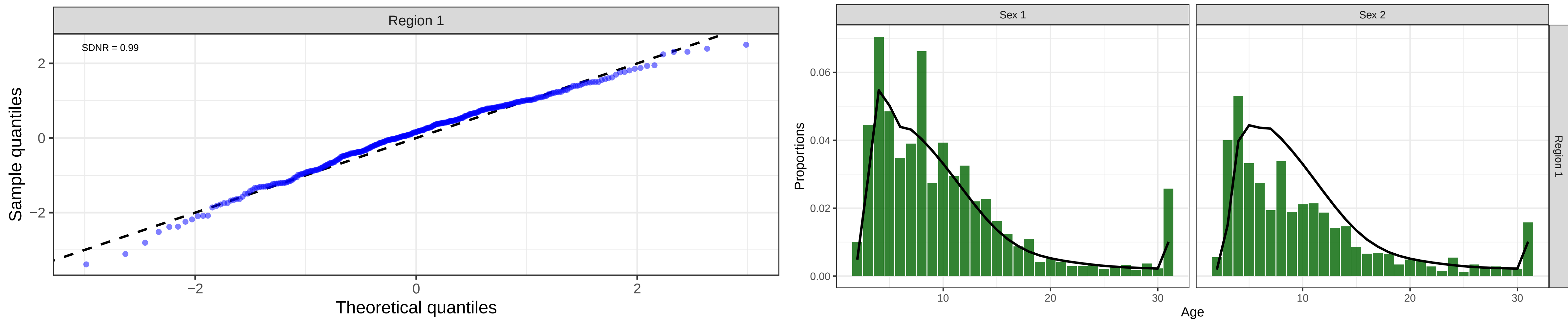
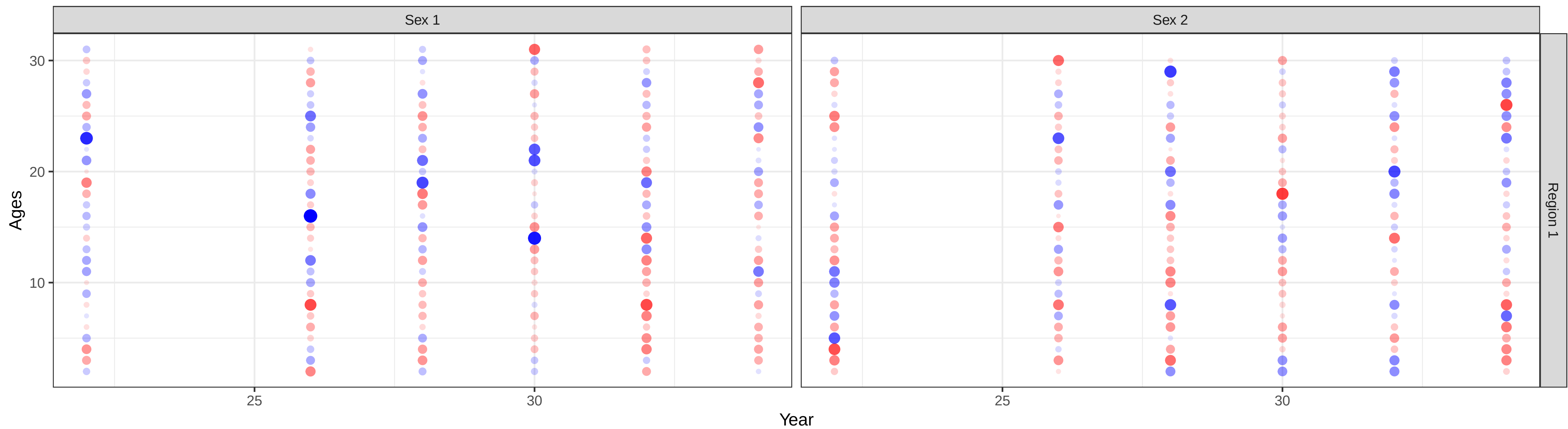




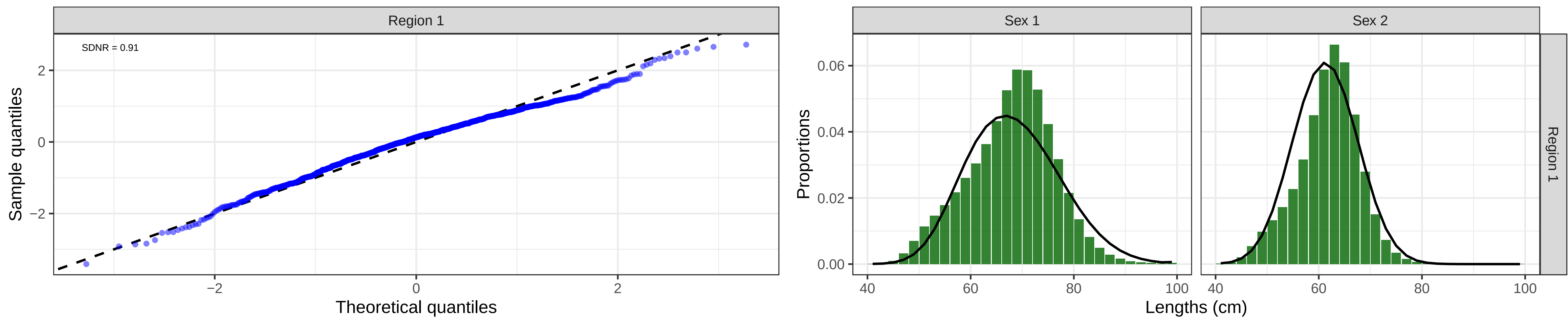
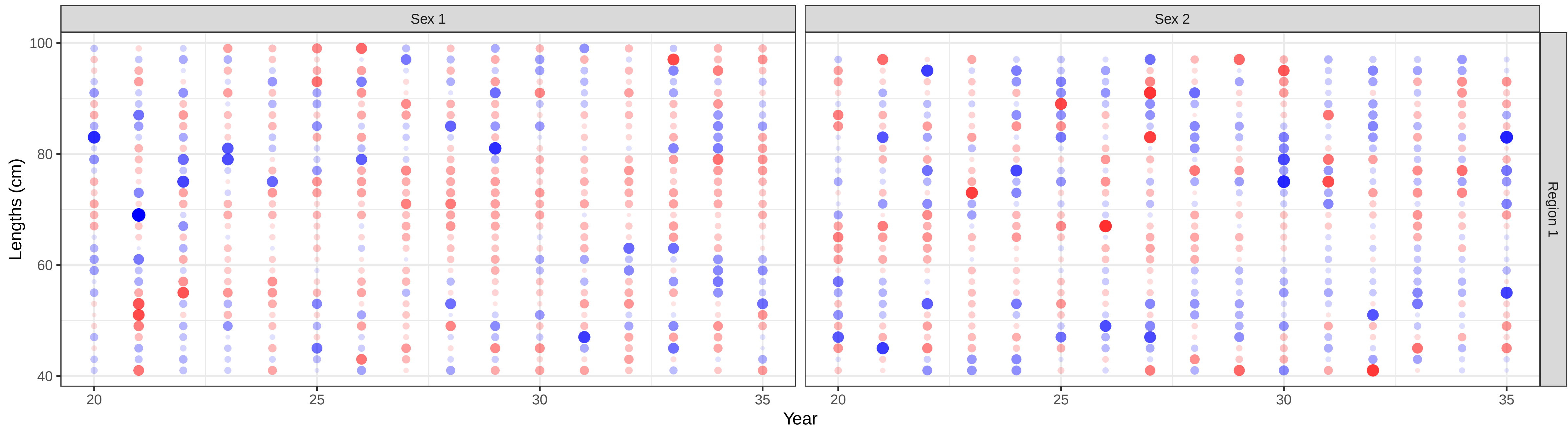
abs(Resid) 1 2 3 Sign ● Neg ● Pos



abs(Resid) 1 2 3 Sign ● Neg ● Pos

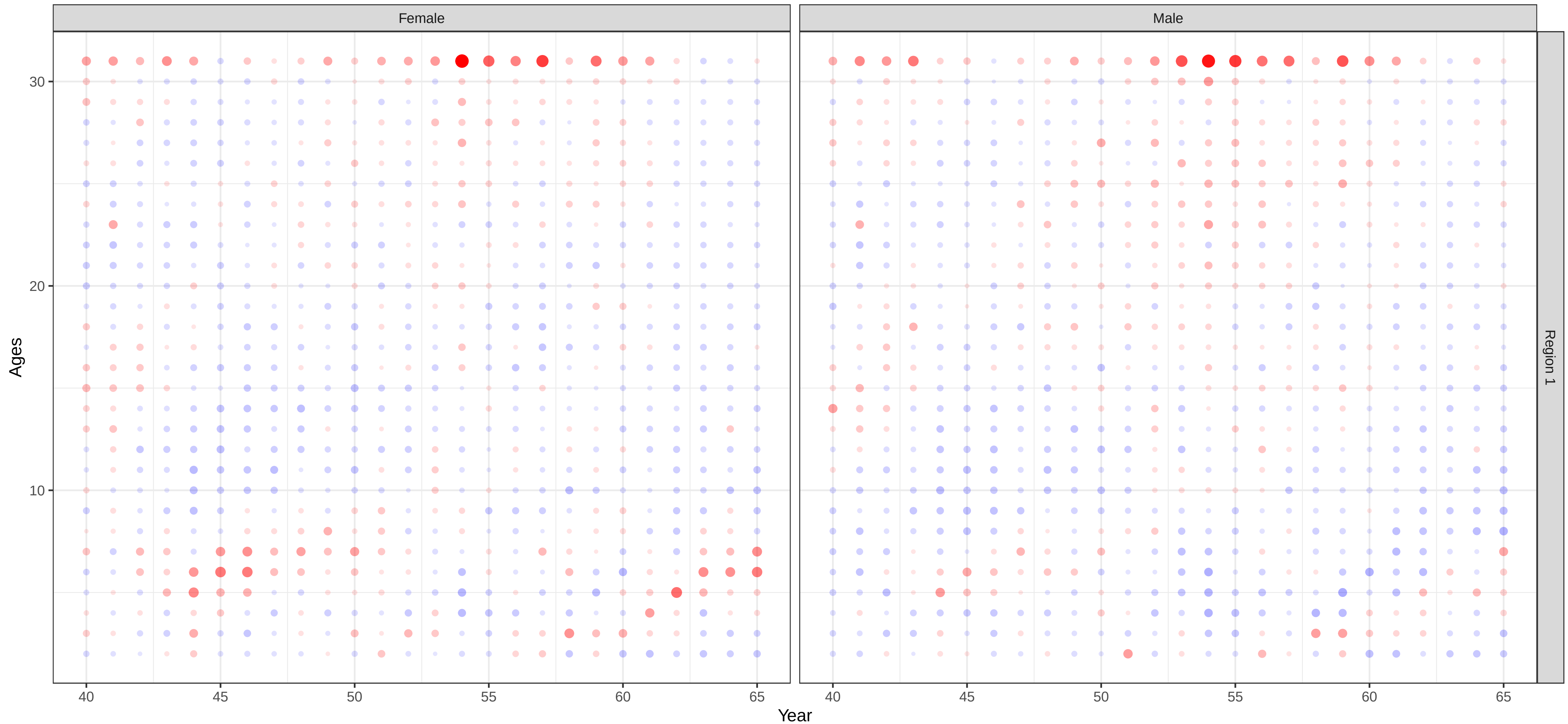


abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos



Fixed Gear Fishery Age Compositions

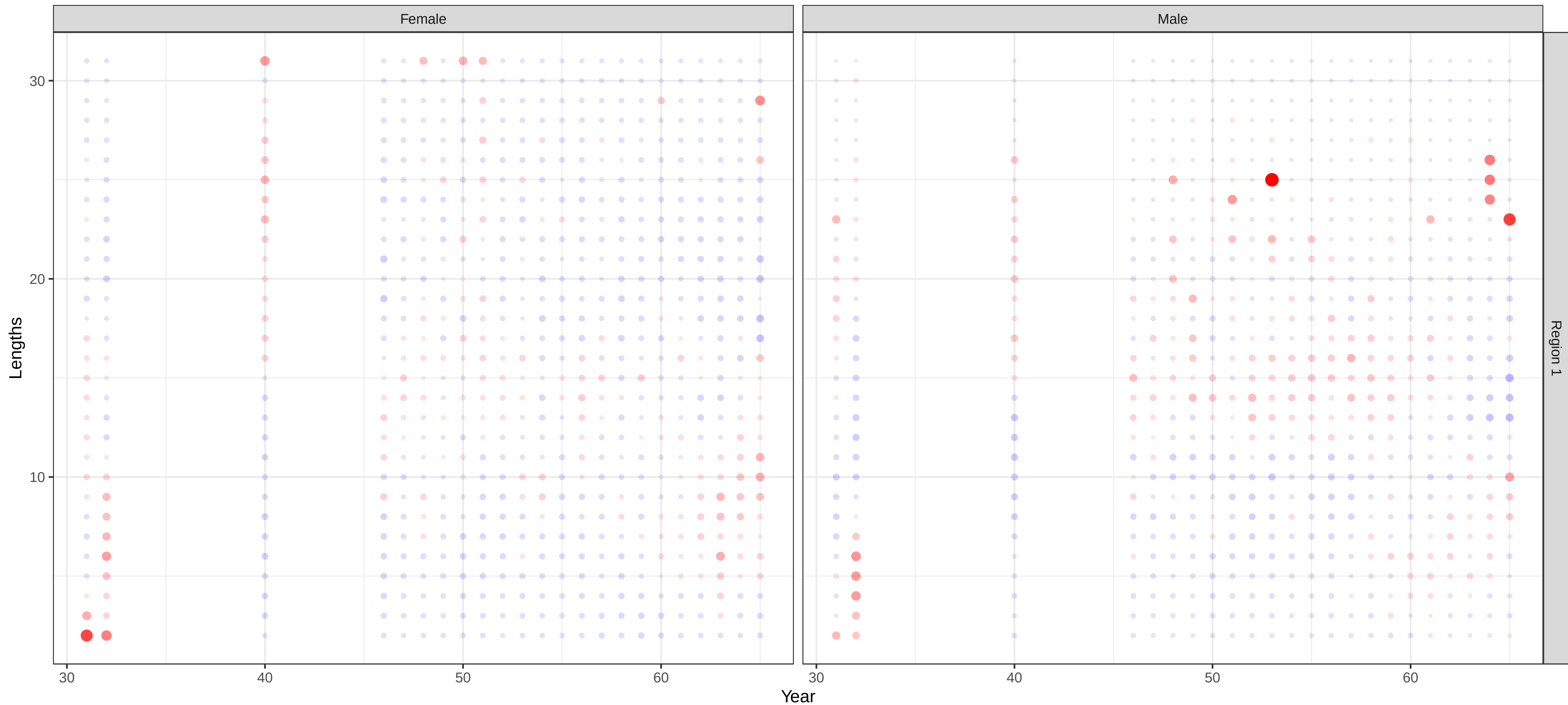
Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



Trawl Fishery Length Compositions

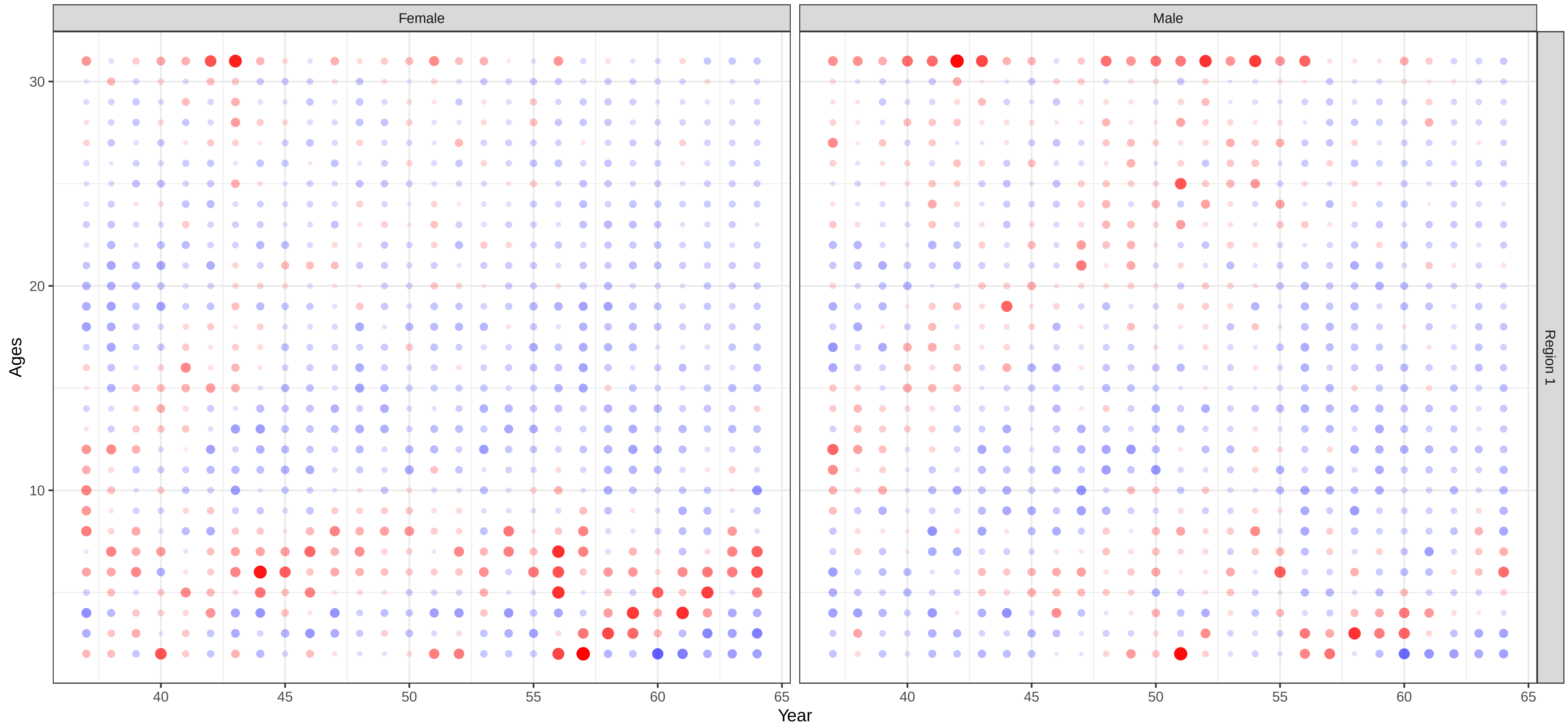
Sign ● Neg ● Pos

abs(Resid) ● 0 ● 2 ● 4
 ● 1 ● 3 ● 5



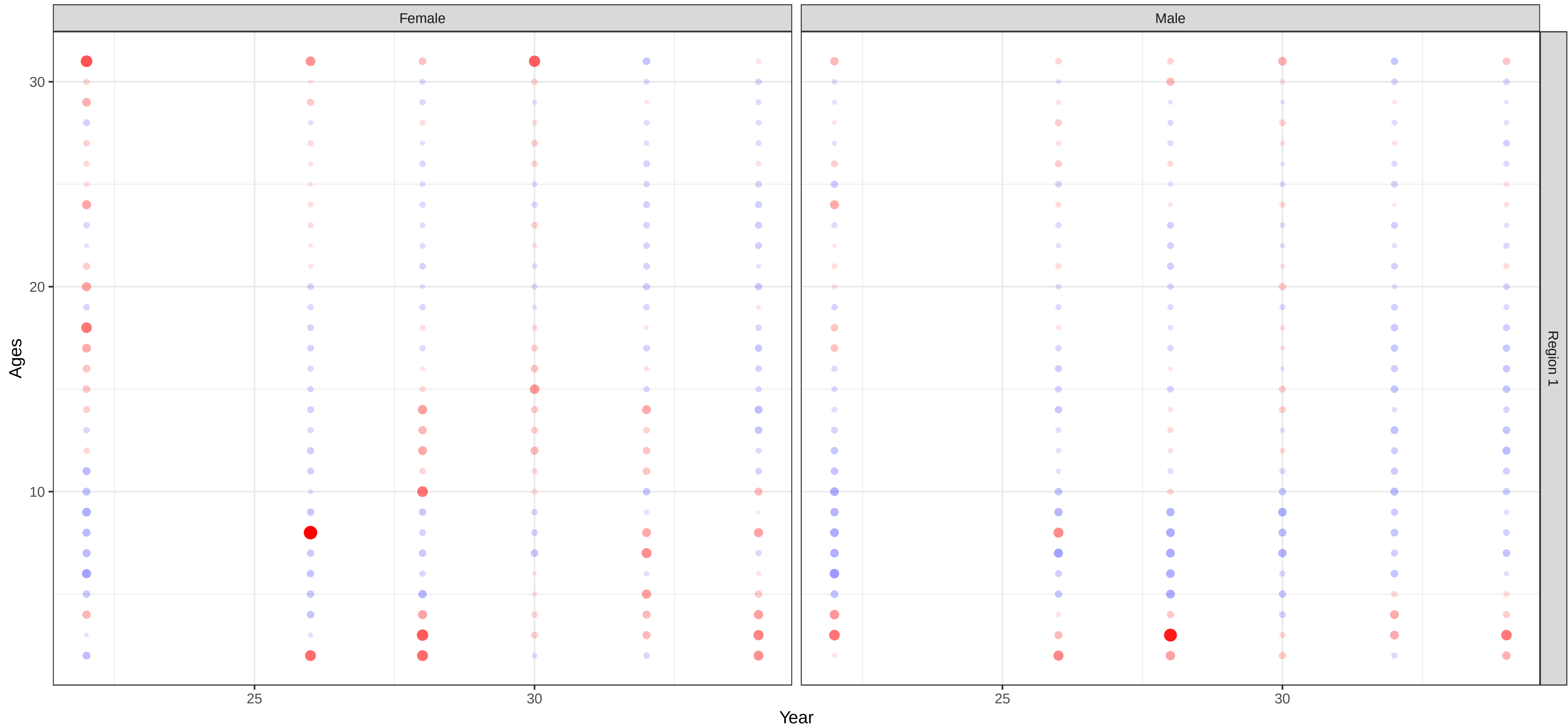
LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0



JPN LLS Age Compositions

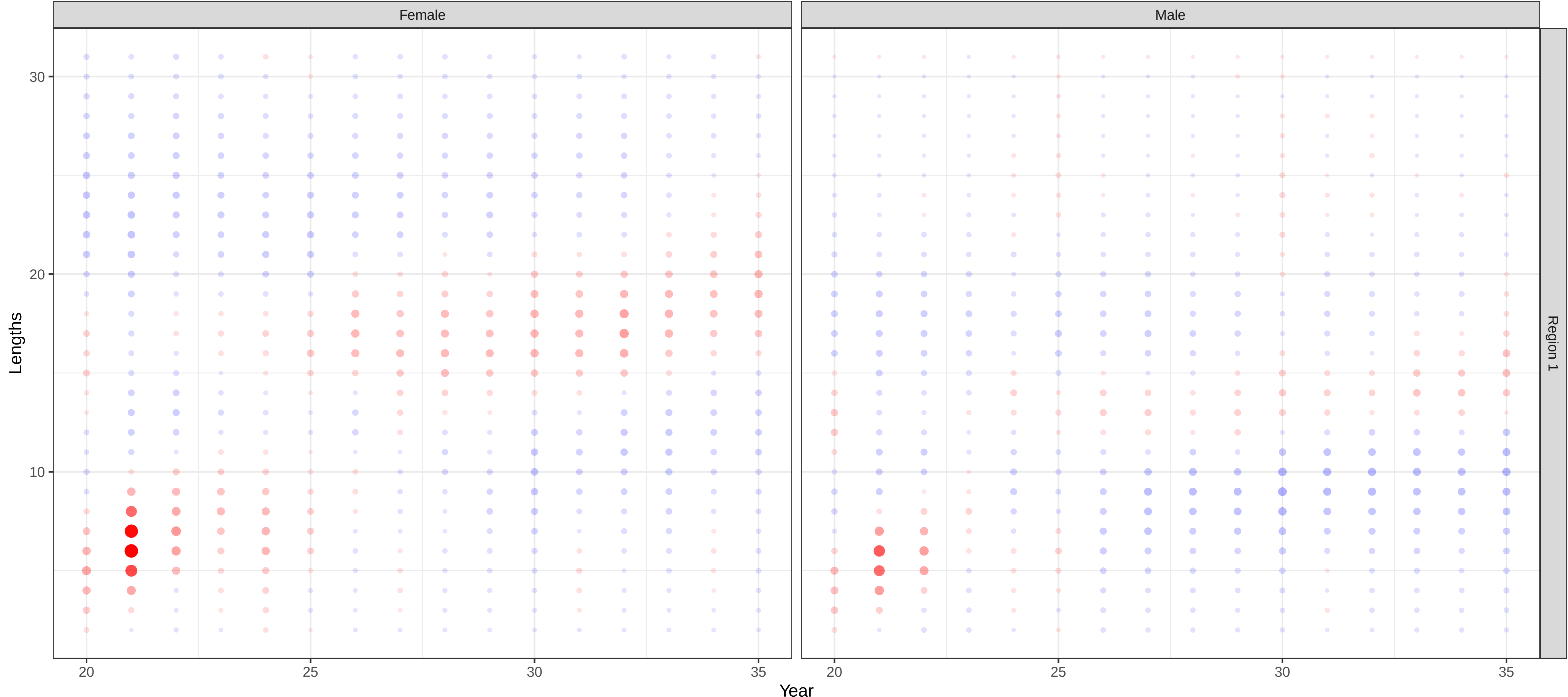
Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0

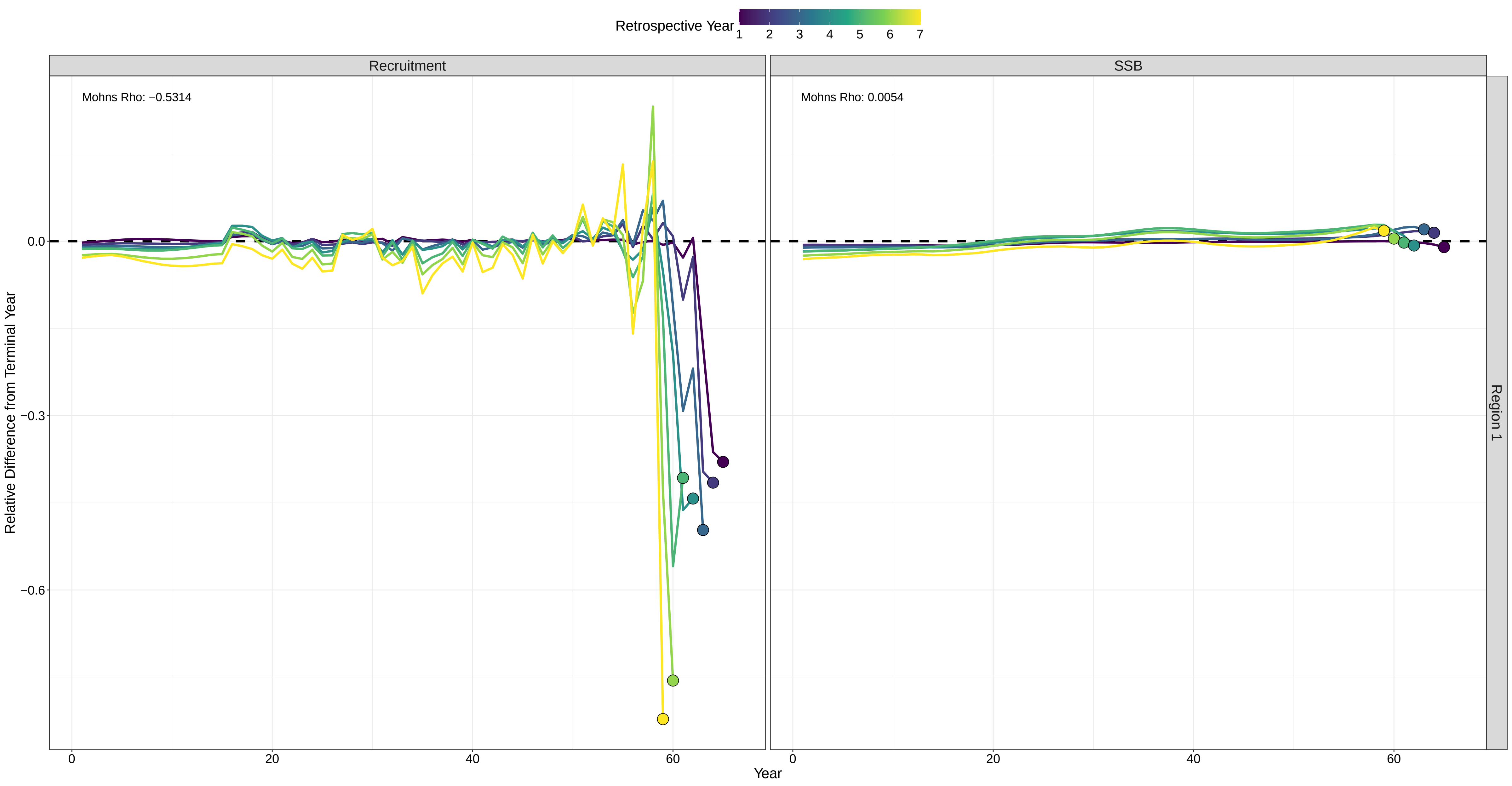


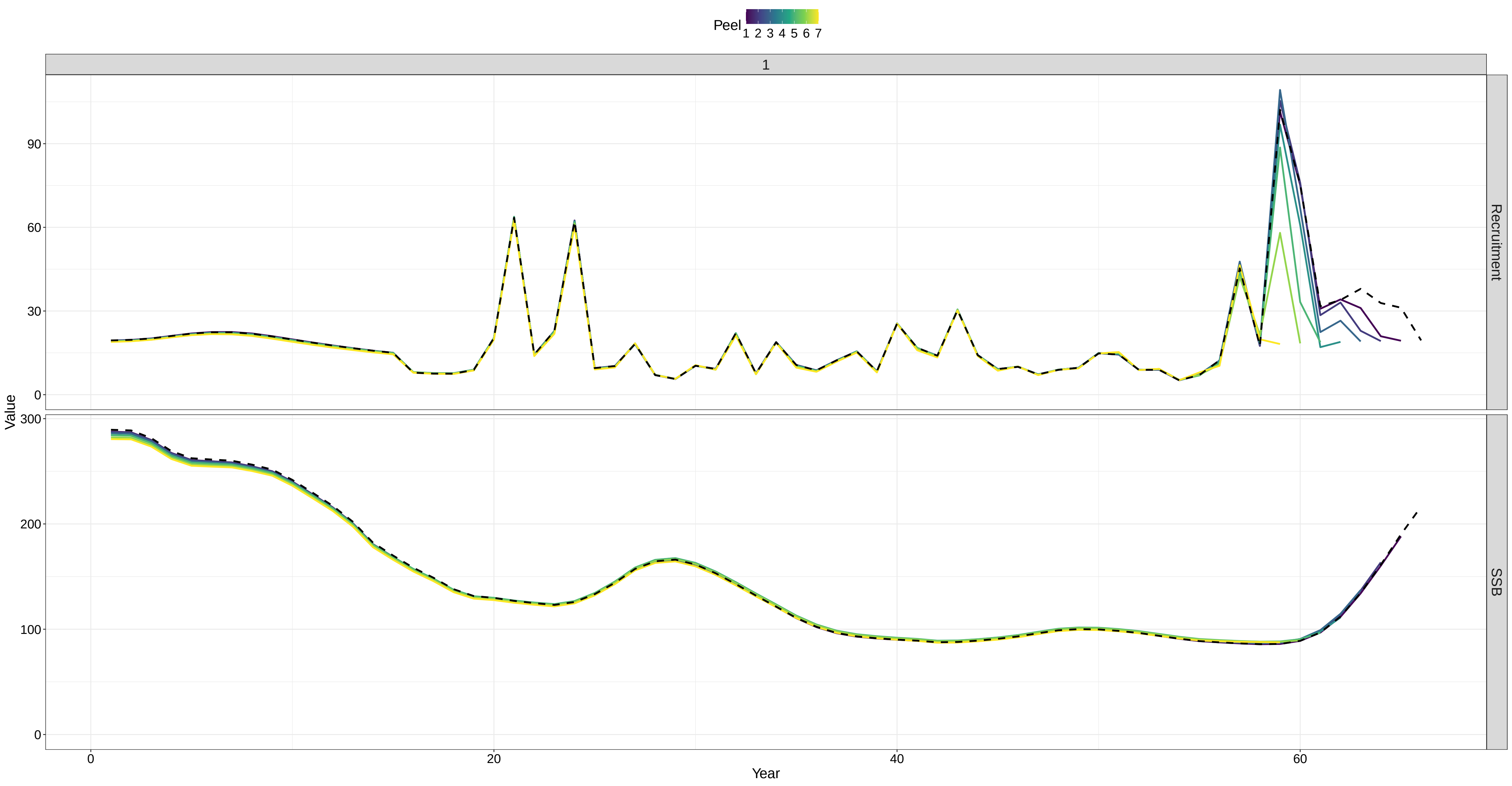
JPN LLS Length Compositions

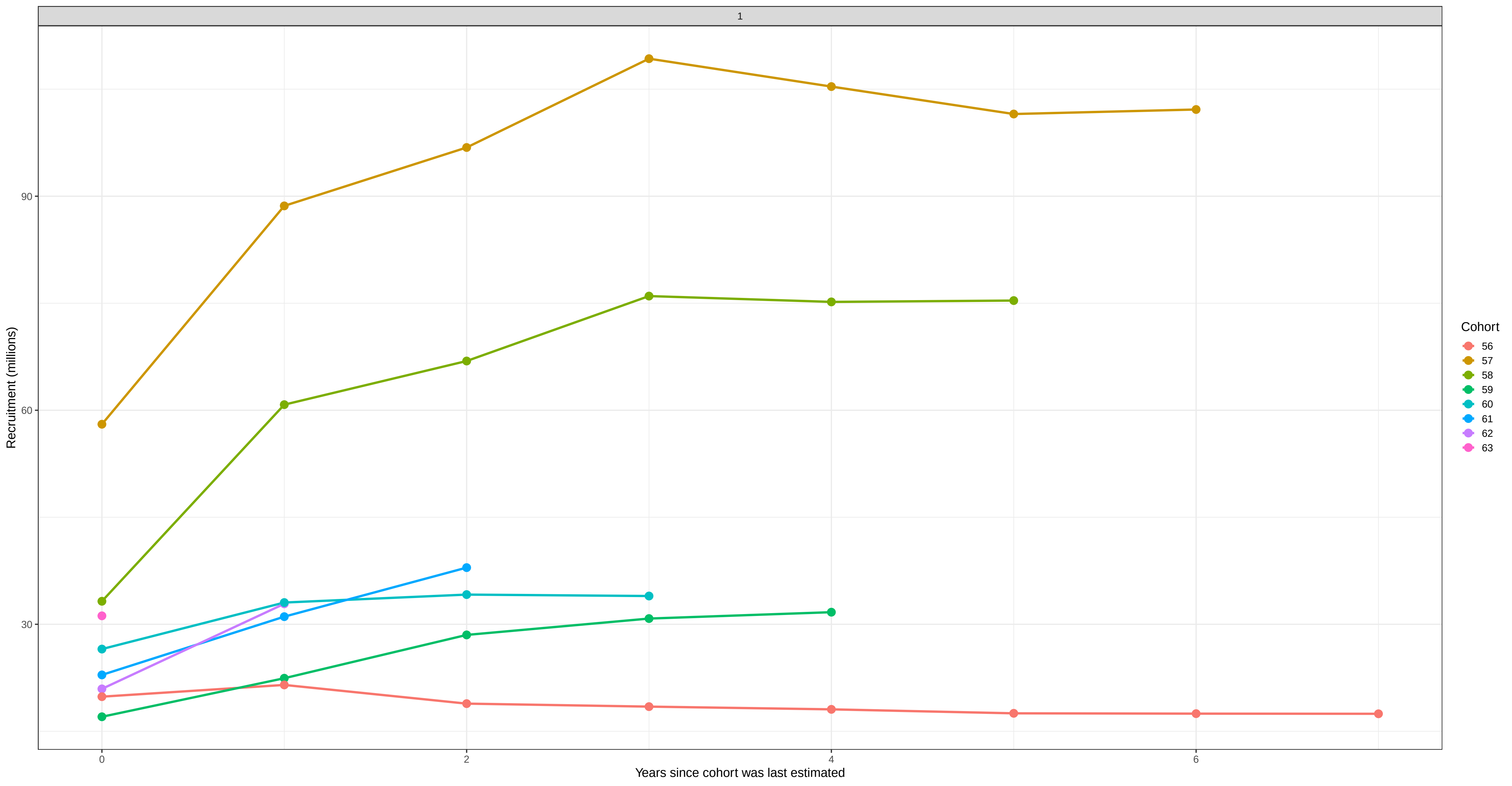
Sign ● Neg ● Pos

abs(Resid) ● 0.0 ● 1.0 ● 2.0
 ● 0.5 ● 1.5 ● 2.5

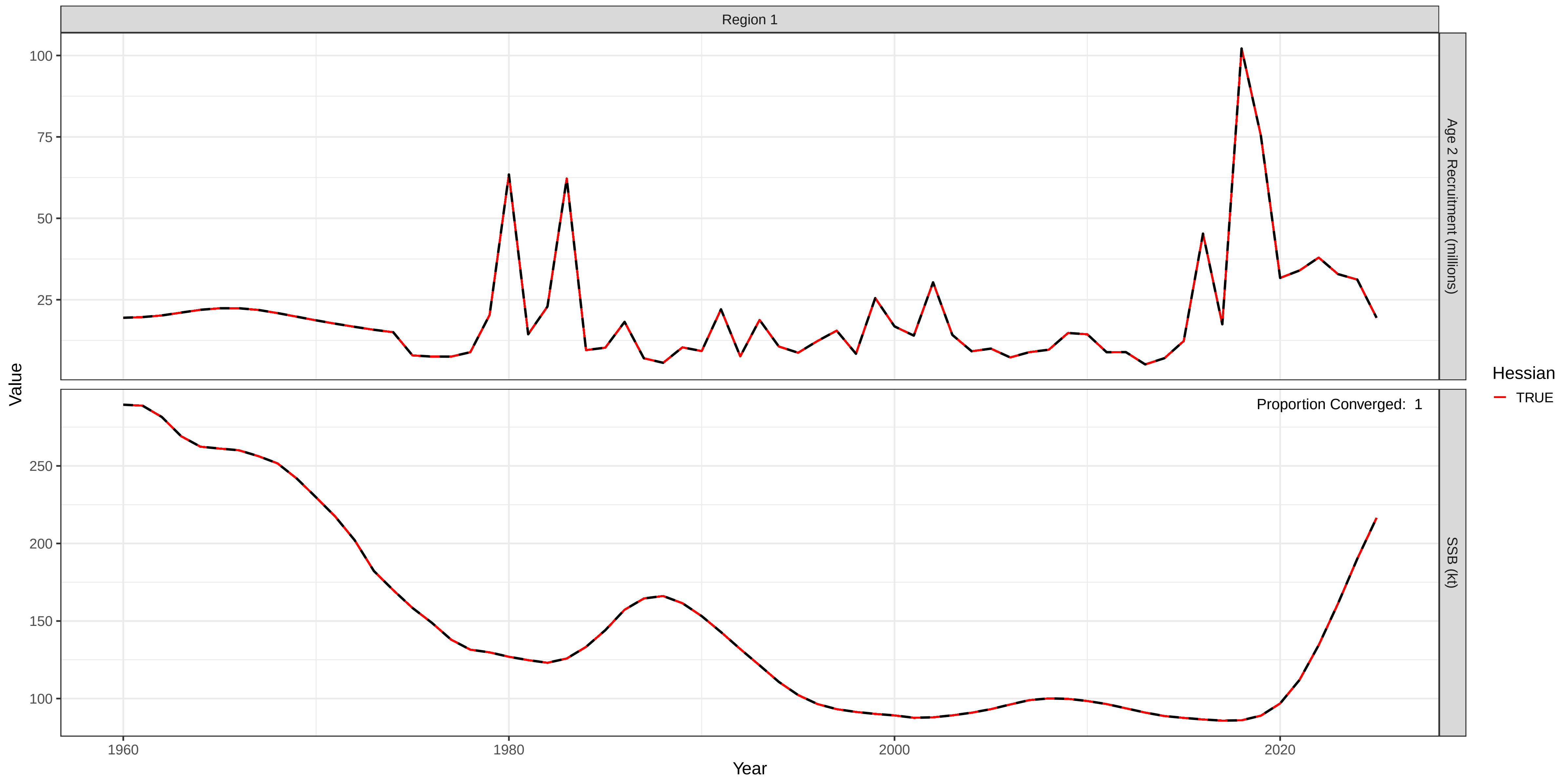




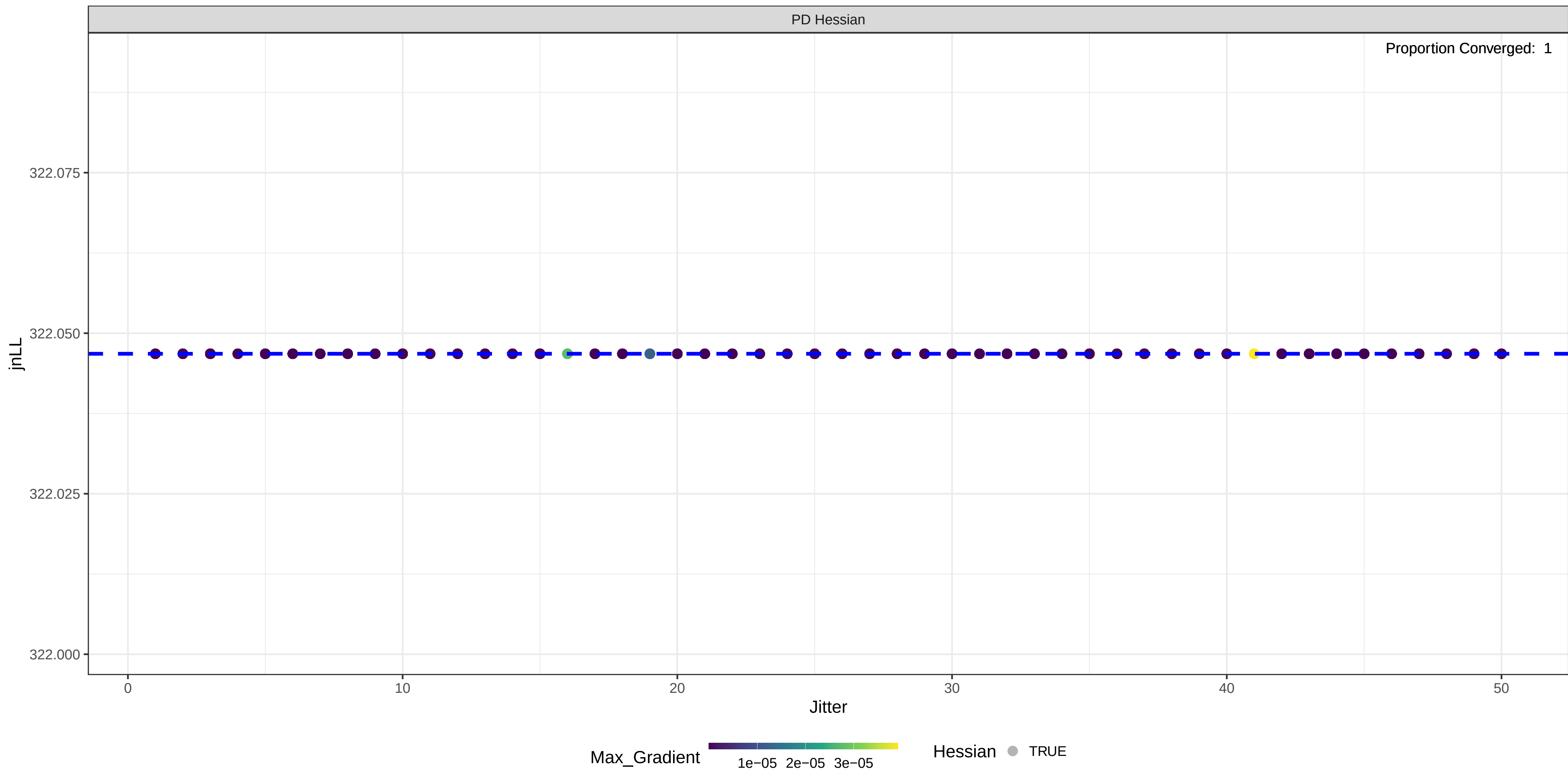




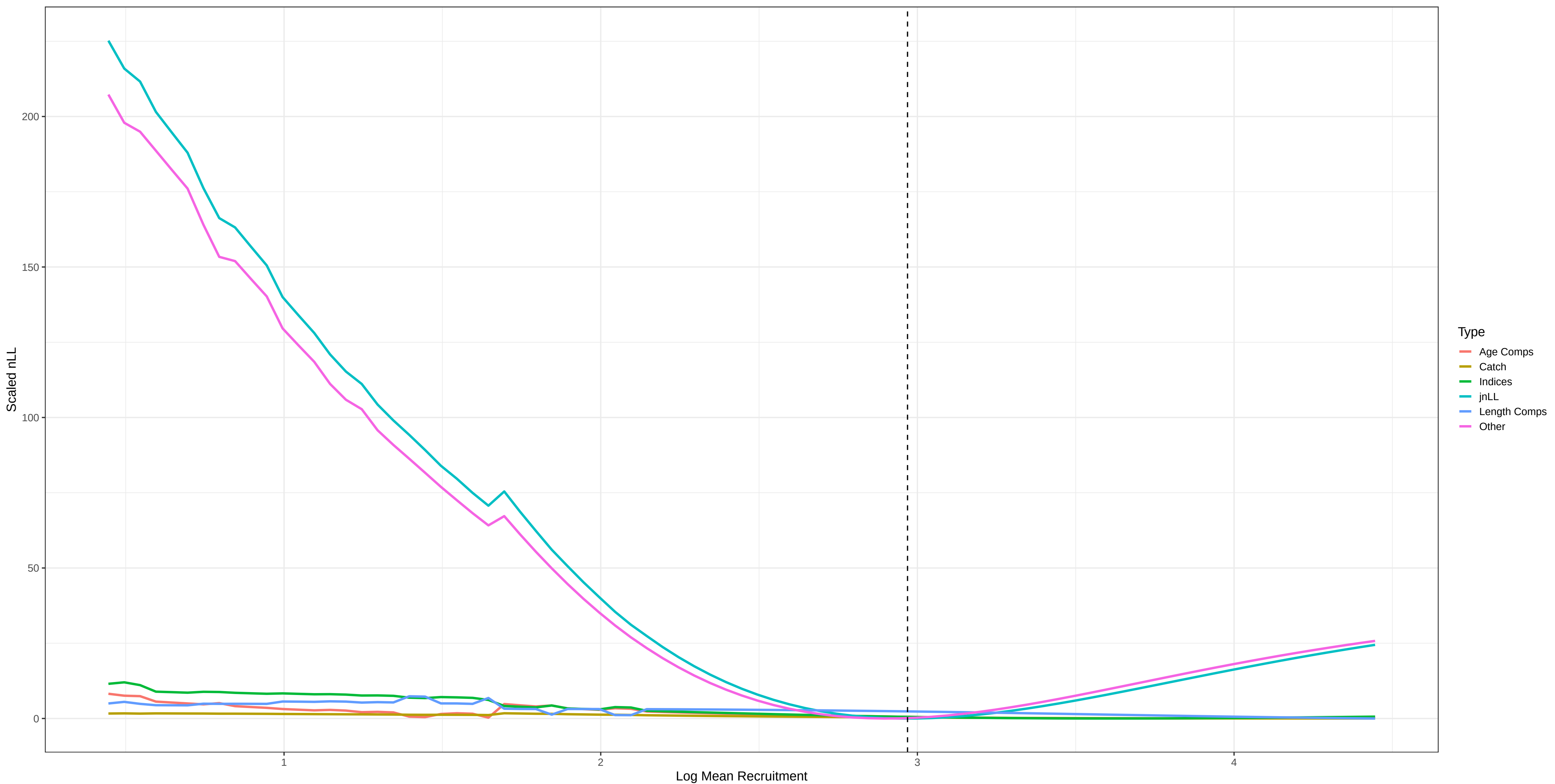
Jitter SSB and Recruitment



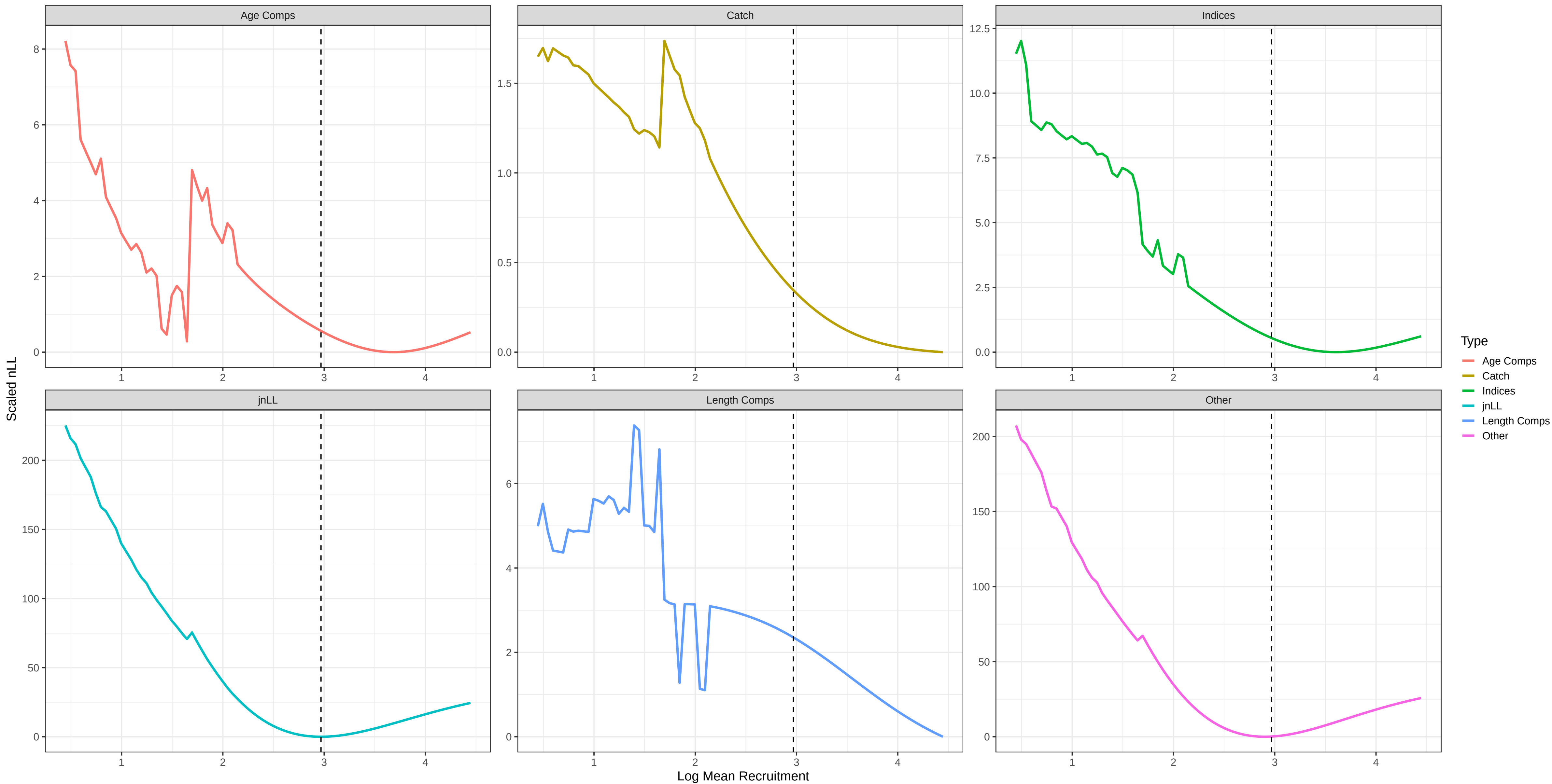
Jitter nLL and Convergence



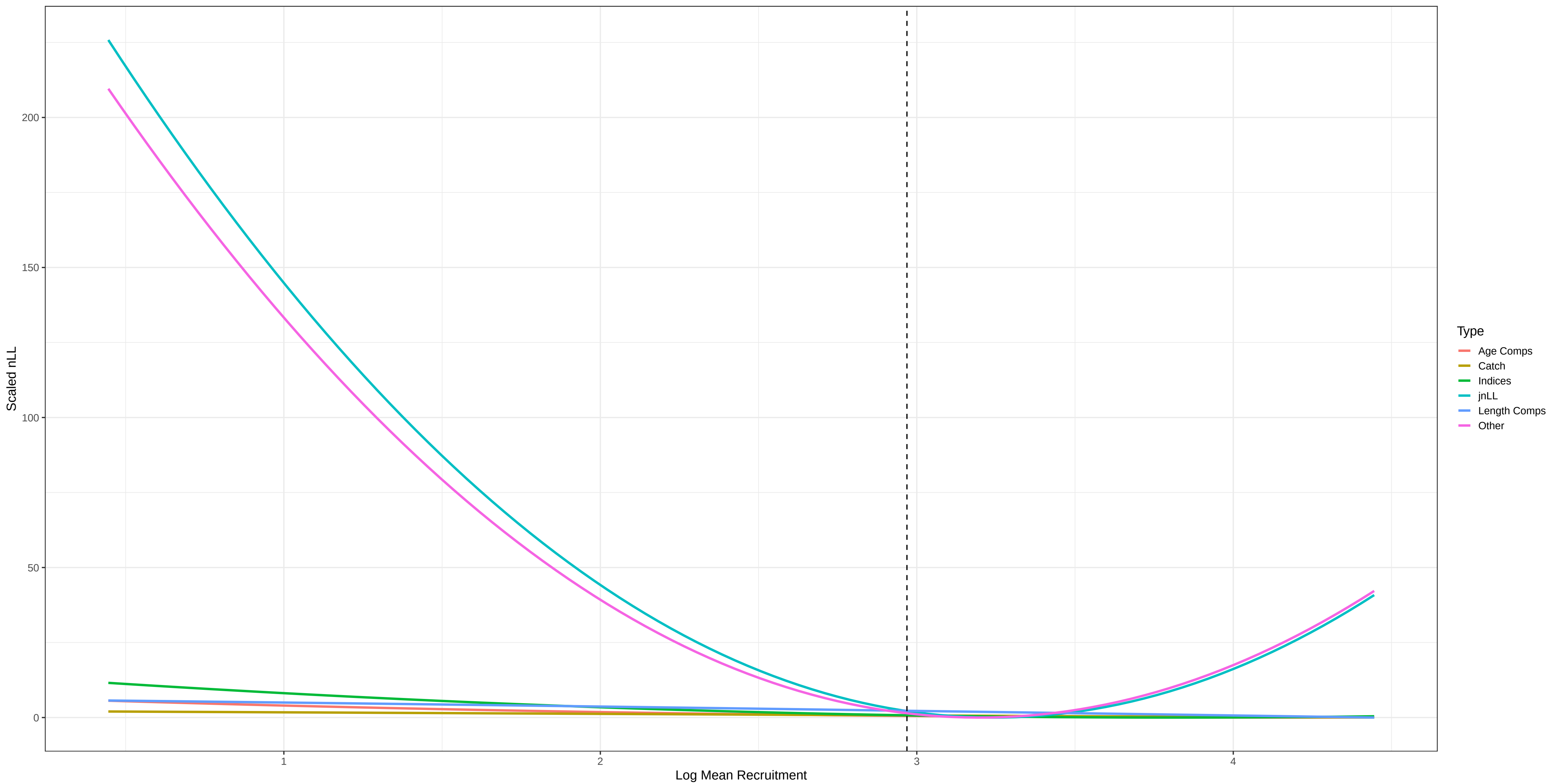
Profile Likelihood R0



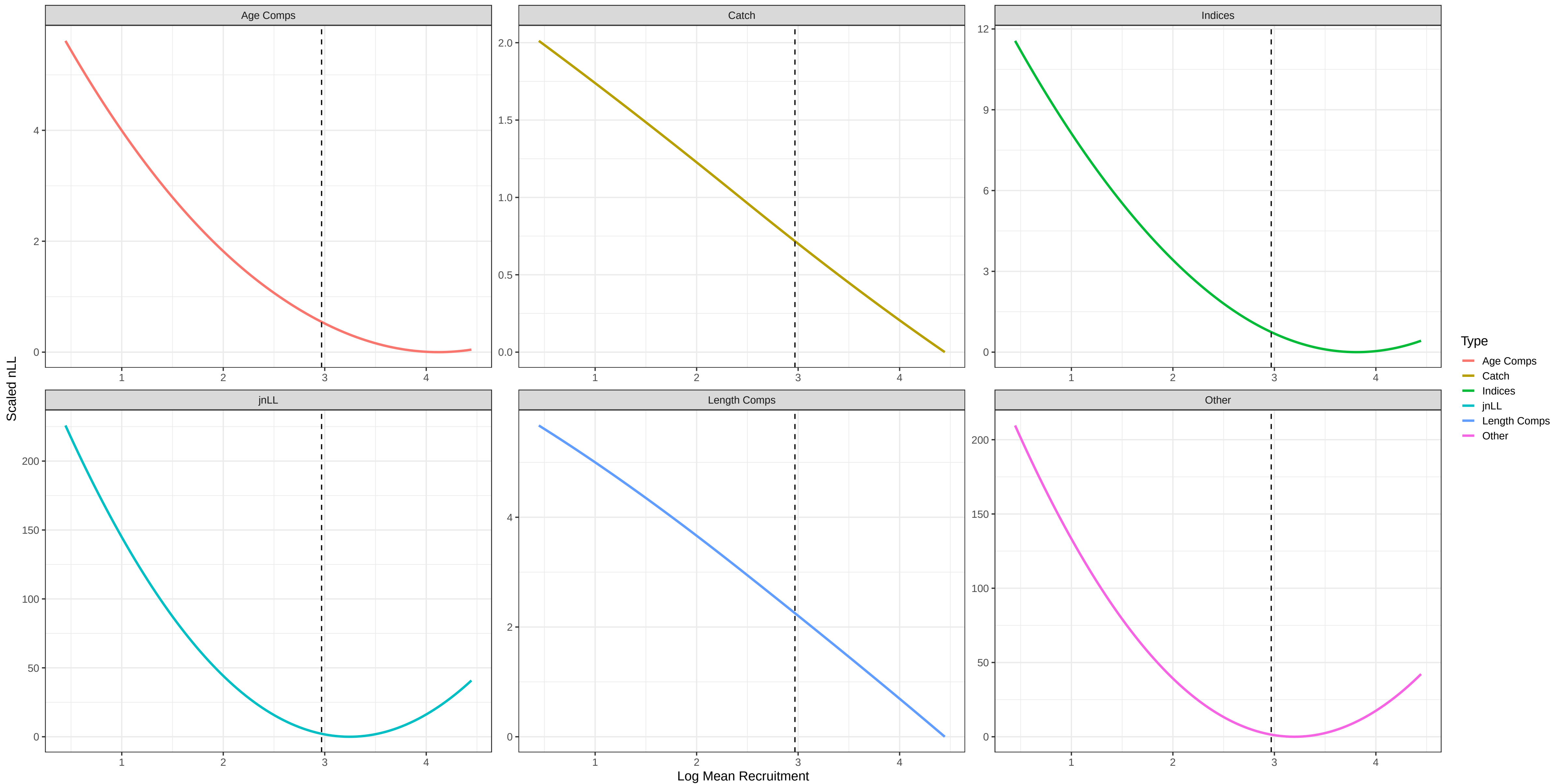
Profile Likelihood R0



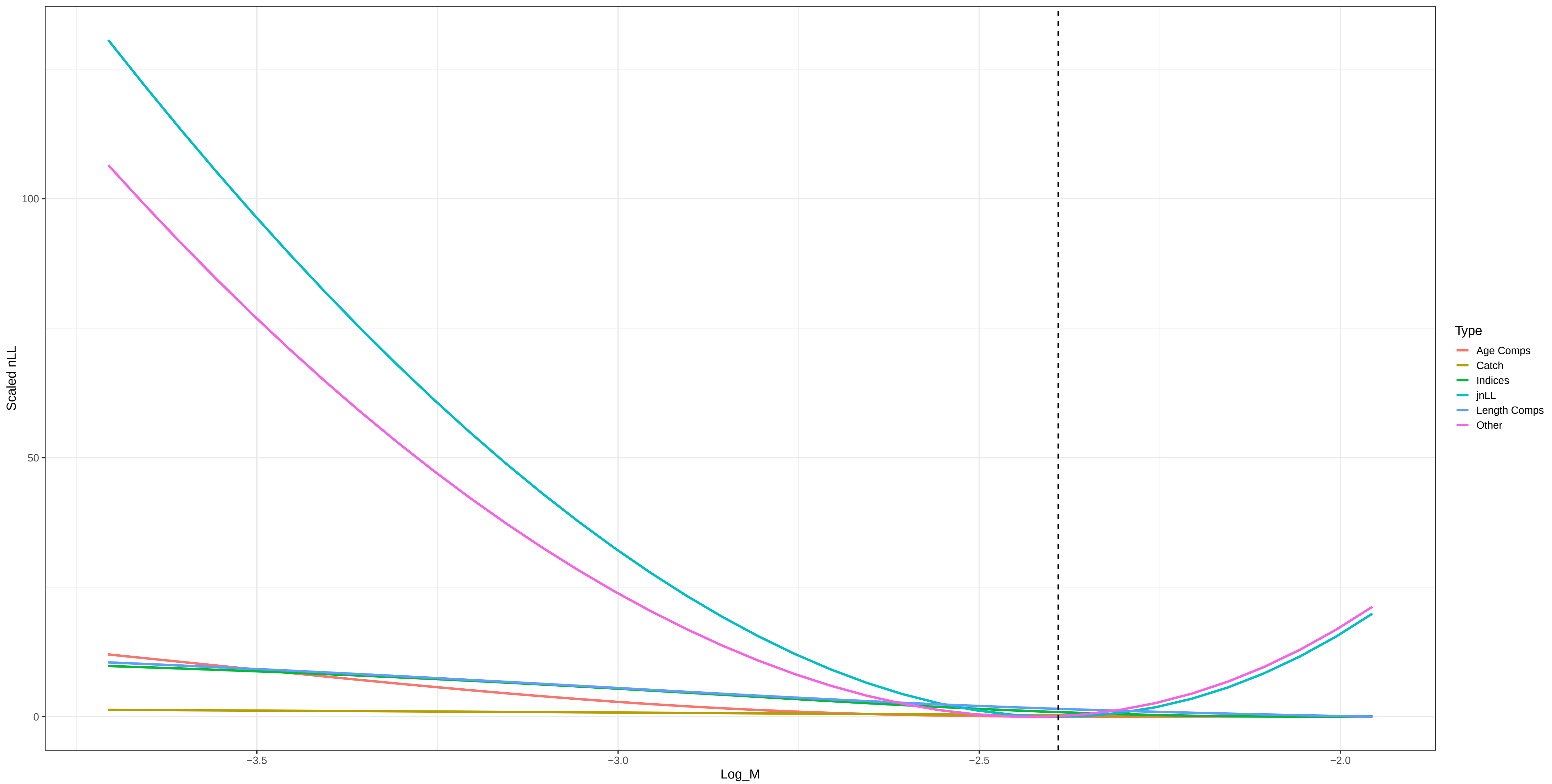
Profile Likelihood R0



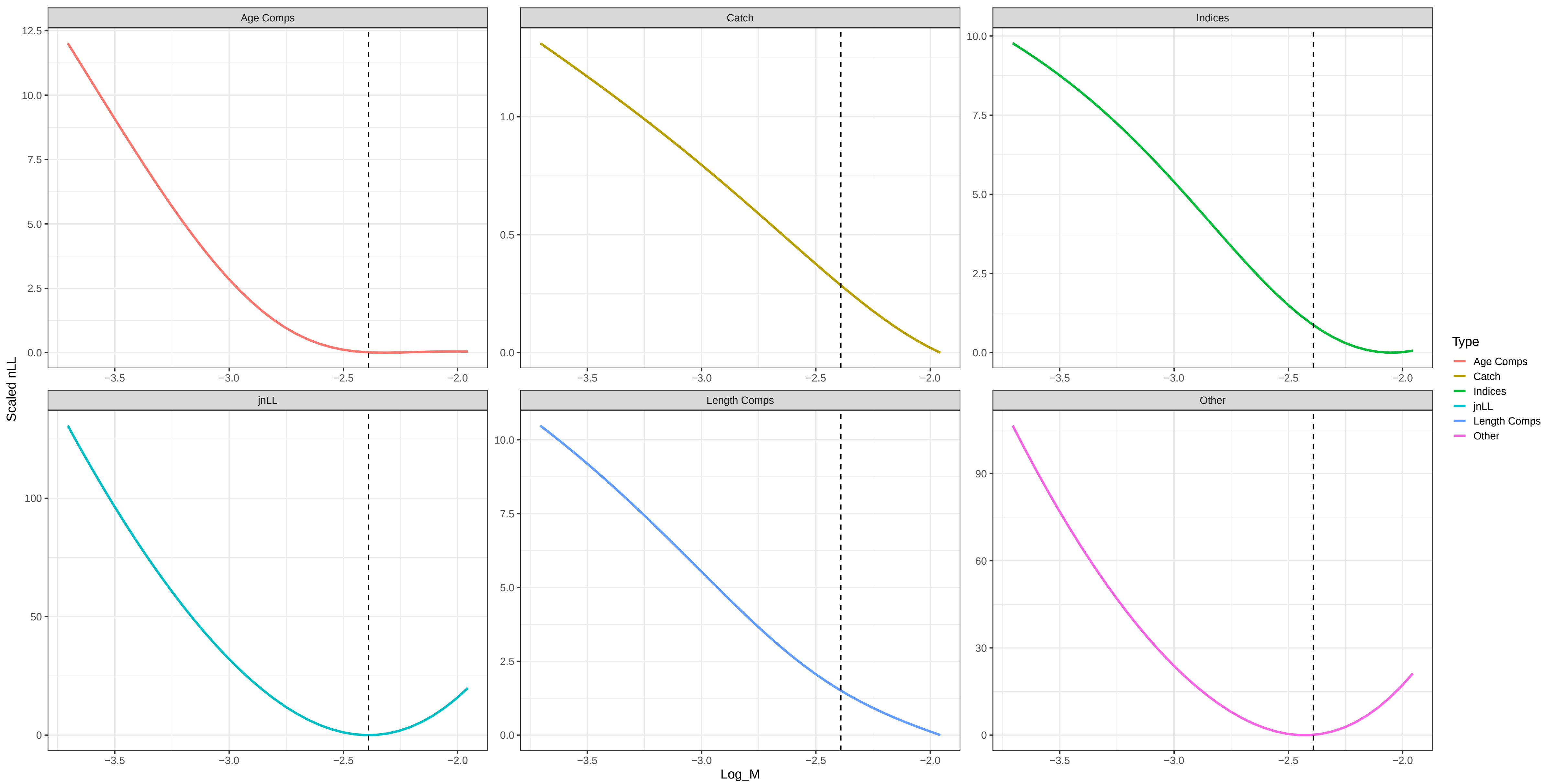
Profile Likelihood R0



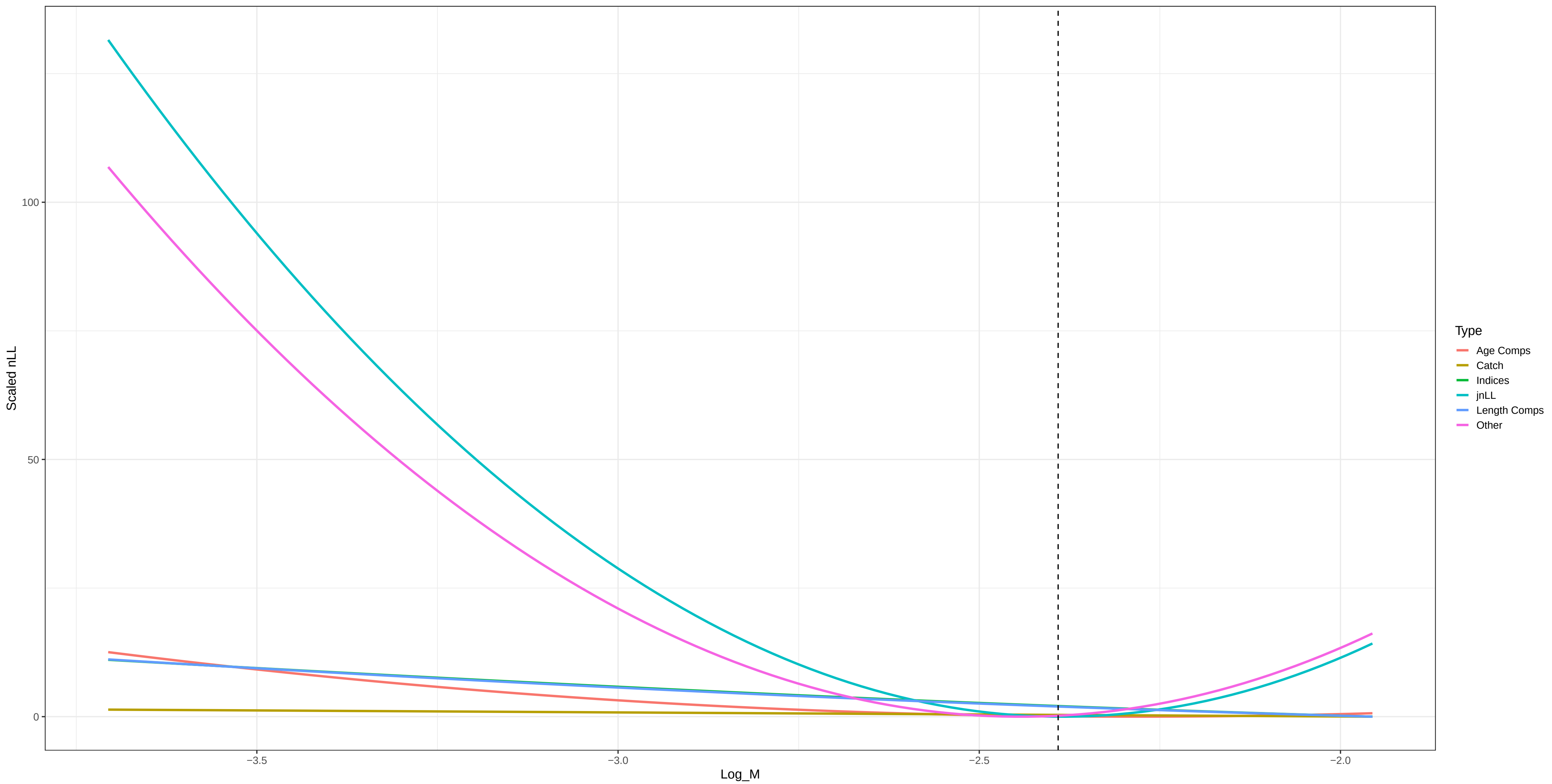
Profile Likelihood Female M



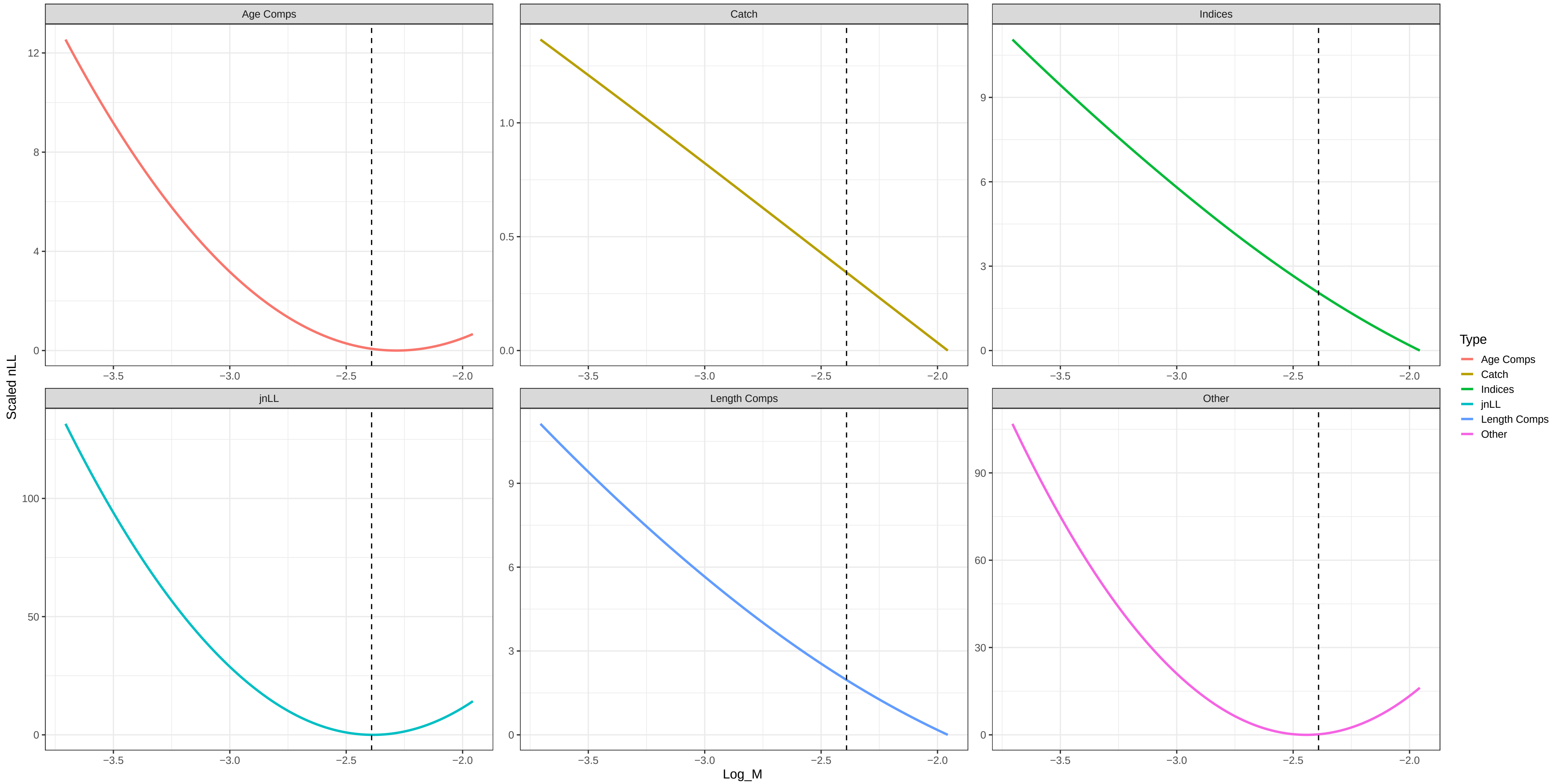
Profile Likelihood Female M



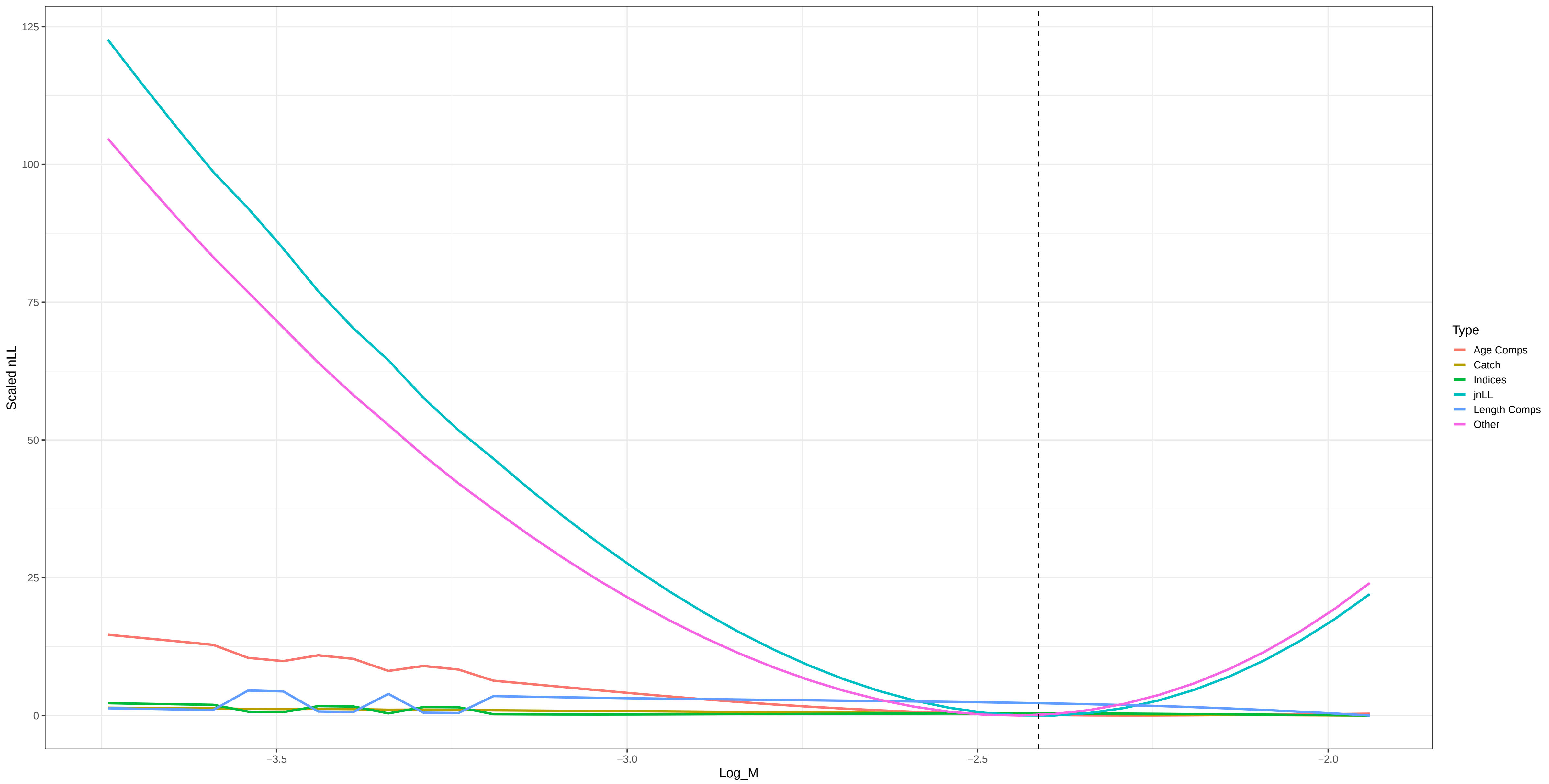
Profile Likelihood Female M



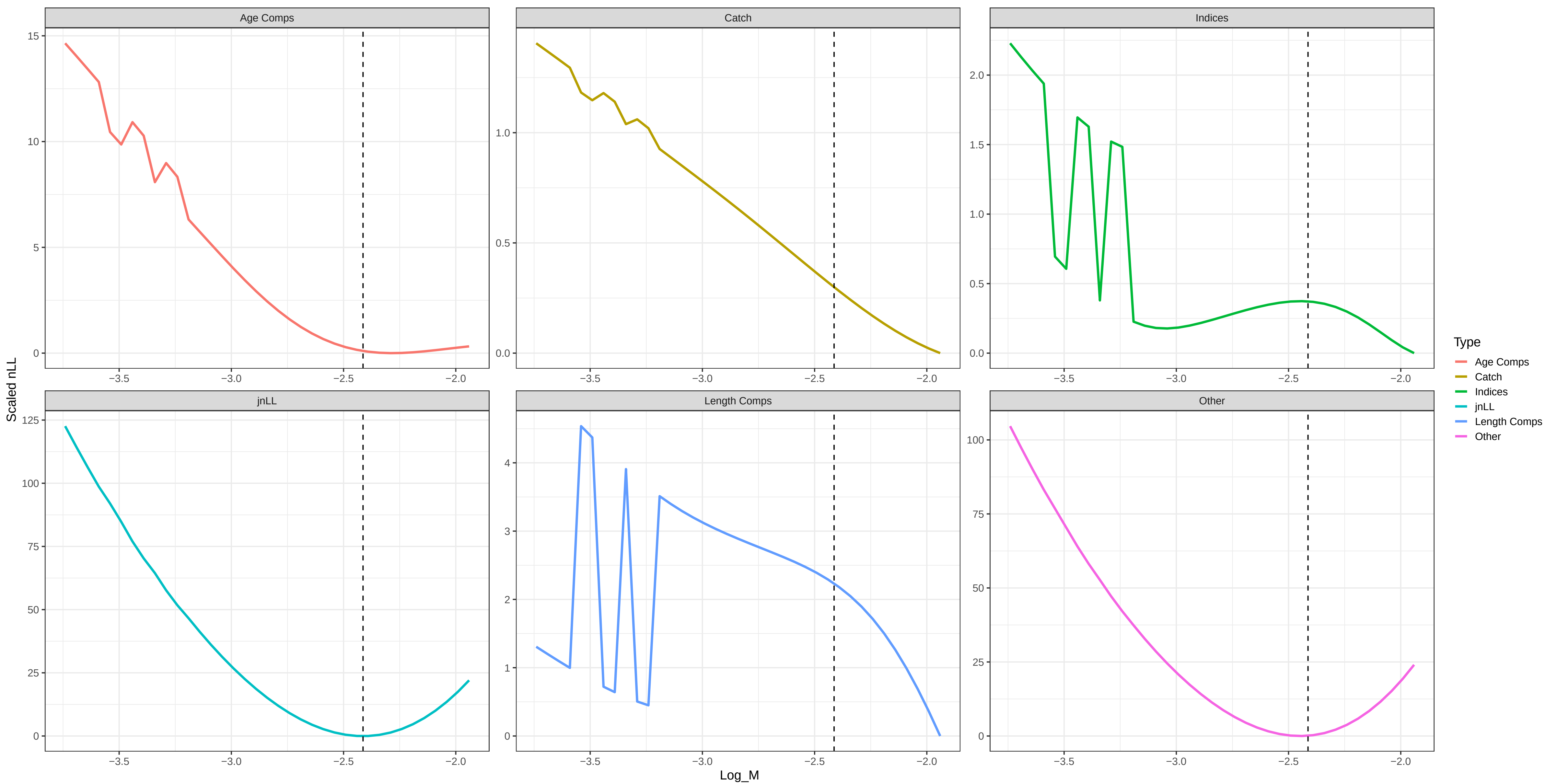
Profile Likelihood Female M



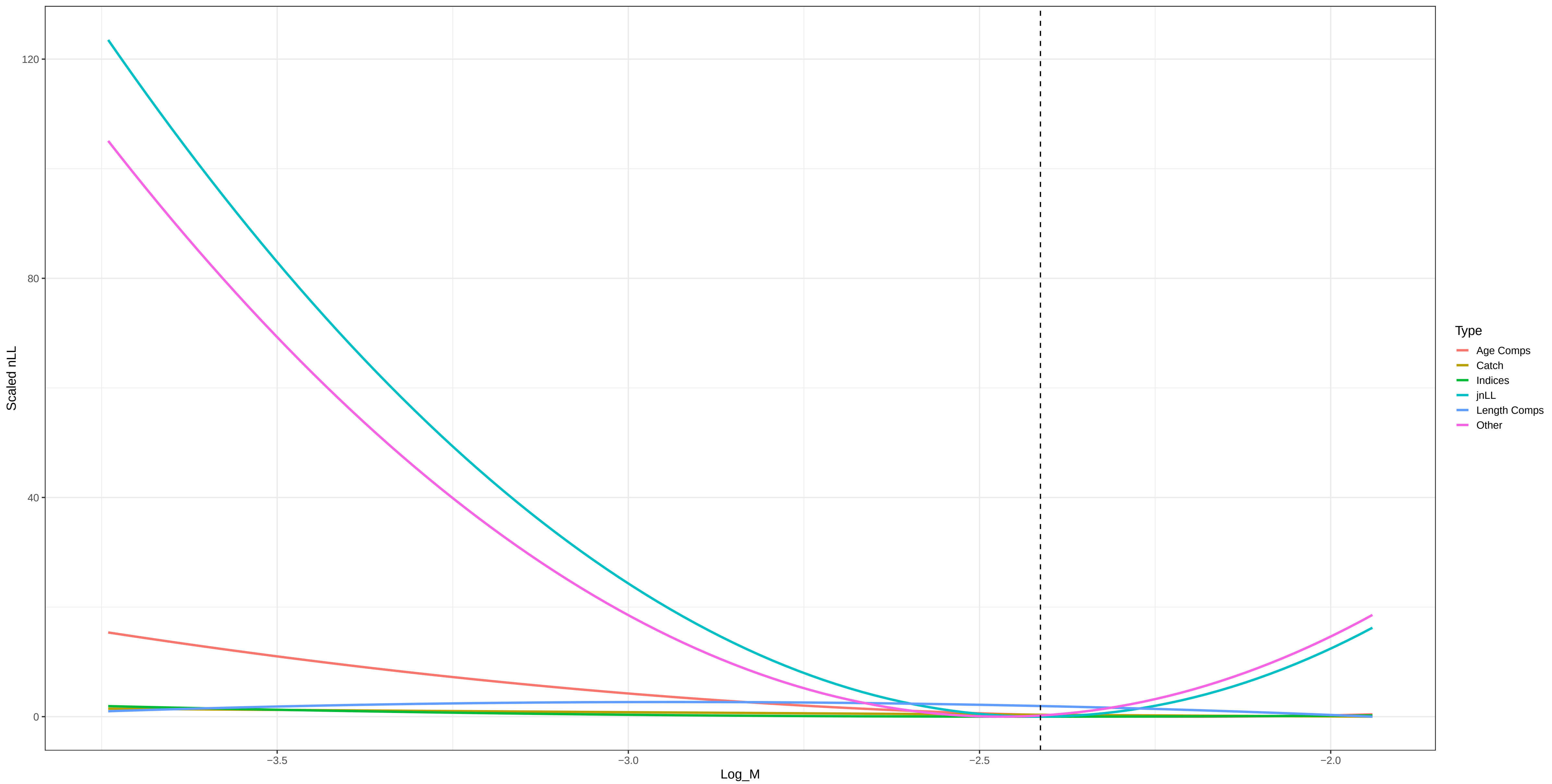
Profile Likelihood Male M



Profile Likelihood Male M



Profile Likelihood Male M



Profile Likelihood Male M

